

The Enforcement Bottleneck: Judicial Capacity and the Limits of Bankruptcy Reform

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July 14, 2026

Abstract

Efficient capital allocation requires both a credible threat of enforcement and enough information to identify productive ones. Well-enforced bankruptcy reforms typically strengthen the first while leaving the second untouched. I study India’s 2016 national bankruptcy reform, which strengthened creditor rights, using a triple difference research design that exploits variation in time, local judicial capacity (district-level judge-to-population ratios), and firms’ pre-reform default risk or capital efficiency. In districts with greater judicial capacity, the reform produced a credit-disciplining effect: high-default-risk firms borrowed less, held more cash, and contracted their operations relative to low-risk firms. However, it did not raise relative borrowing by more productive firms—whose productivity is poorly captured by the credit ratings lenders rely on.

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1 Introduction

Efficiently allocating capital among firms in developing countries requires two things at once: the ability to withdraw credit from unproductive firms, and the ability to direct it toward more productive ones (Hsieh and Klenow 2009; Banerjee and Duflo 2014; Buera et al. 2011). The first is limited by an enforcement friction: lenders cannot credibly recover capital from defaulting borrowers in the presence of weak creditor rights and an inefficient last-mile bankruptcy enforcement (La Porta et al. 1998; Djankov et al. 2007; Gormley et al. 2018). The second is limited by an information friction: lenders cannot readily observe which firms are most productive and are able to repay. A large literature has studied bankruptcy reforms through the first lens, with mixed results (Vig 2013; Ponticelli and Alencar 2016; Breza and Liberman 2017; Closset et al. 2023) that often hinge on local enforcement capacity (Giné and Love 2010; Chemin 2020; Alok et al. 2022). This paper asks whether relaxing the enforcement friction is sufficient to improve allocation, or whether the information friction continues to bind.

I examine this question within the context of one of the largest legislative and policy reforms in bankruptcy protection in India (Insolvency and Bankruptcy Reform or IBC, 2016). This setting provides an ideal empirical design to test the limits of legal reforms on subsequent firm-level borrowing and outcomes for three reasons. First, the reform was a national shock that shifted power from borrowers to creditors. Second, there is substantial variation in local judicial capacity - measured by granular, district-level judge-to-population ratios - that affects enforcement processes involved in the recovery of capital and subsequent allocation of credit. Third, firms in India face credit constraints to be on the productivity frontier (Hsieh and Klenow 2009; Banerjee and Duflo 2014; Breza and Kinnan 2021) and observable signals of their repayment ability could affect their borrowing. This allows me to use a triple difference (DDD) research design that exploits variation in the timing of the reform with predetermined spatial variation in district court capacities and predetermined variation among local firms based on their default risk (measured by their credit rating) or capital efficiency (measured as estimated Marginal Revenue Product of Capital or MRPK).¹

I find that in districts with high judicial capacity, the reform leads to credit disciplining

¹MRPK is an imperfect measure of firm's capital efficiency extensively used in the literature, for example, Bau and Matray (2023). Ideally, one would use marginal product of capital (MPK) but because data is mainly available for sales revenue instead of production, MRPK is used as a close proxy. These firm-level characteristics are generated based on data on credit rating and estimated MRPK before 2010. This addresses any endogeneity or anticipation concerns given the reform timeline. District judicial capacity is constructed using judge tenure data from Ash et al. 2025 for the period 2012-2016.

effect: high-default risk firms reduce their borrowing and increase short-run liquidity. However, I find no evidence that the reform redirected credit towards high capital efficiency firms, even in high judicial capacity districts. These results suggest that while judicial capacity could reduce credit allocation based on an observable and incentivized dimension of credit risk such as credit rating (Chemin 2010; Ponticelli and Alencar 2016; Li and Ponticelli 2022; Müller 2022), it did not, on its own, redirect credit towards more productive firms.

In India, local district courts played a direct role in bankruptcy process under the Provincial Insolvency Act, 1920, which the IBC 2016 reform superseded. Practically, the reform strengthened lender rights through a greater focus on asset restructuring and less on liquidation (in contrast to Chapters 11 and 7 of the US bankruptcy code, respectively).² The reform included a new legislative act and a series of policies by India’s central bank - the Reserve Bank of India (RBI) - such as the 2015 Asset Quality Review (AQR) and 2018 circular that aimed to improve the efficacy of the new law by declaring bad debts promptly. However, in practice, district courts continue to play a complementary role, including in the execution of judicial orders, injunctions against asset liquidation, and challenges to orders passed by specialized bankruptcy courts (NCLT or DRTs). 44% of liquidation orders under IBC have been stuck in implementation due to multiple ongoing litigation challenging these orders (Insolvency and Bankruptcy Board of India 2026).³ As a result, local judicial capacity in district courts could be critical with respect to the *last mile enforcement* of national-level reforms, which in turn could affect the functioning of financial markets.

The assumption for causal identification using triple difference research design is that the outcomes of firms by default risk (or MRPK) trend similarly within subgroups of districts with higher or lower judicial capacities in the absence of the bankruptcy reform. This design relaxes the more stringent parallel trends assumptions but requires that high and low-default risk firms (or high and low MRPK firms) trend similarly within high or low judicial capacity

²This was emphasized by Dr. M.S.Sahoo, the first chairman of the Insolvency and Bankruptcy Board of India in <https://timesofindia.indiatimes.com/business/india-business/insolvency-laws-aim-is-biz-rejig-not-recovery/articleshow/84125959.cms>. Quantitatively, about 58% of the concluded bankruptcy cases under IBC have resolution favoring restructuring compared to 42% recommended for liquidation. 44% of liquidation orders remain unenforced due to asset valuation disputes, asset auction issues, and legal proceedings in various courts. In contrast, 100% of the restructured firms were “rescued”: 35% found a new buyer and the remaining 65% settled their default with their lender at various stages of the resolution process (Insolvency and Bankruptcy Board of India 2026).

³This is also consistent with the law’s focus on restructuring rather than liquidation. A classic case study is the example of the first liquidation order under IBC, 2016 by India’s bankruptcy court (*VNR Infrastructures Limited v. State Bank of India & Ors*, CP (IB) No. 12/10/HDB/2017). This liquidation order continues to be mired in litigation due to challenges against liquidating one of the properties of the said bankrupt firm (*Leo Edibles and Fats Limited v. Tax Recovery Officer (Central), Income Tax Department, Hyderabad and Others.*, 2018 (3) TMI 1528 / W.P. No. 8560 of 2018).

districts even if they do not trend similarly on average. Since the bankruptcy reform was a one-time national-level reform, applicable to all districts at the same time, it allays any concerns arising from staggered rollout designs ([Ortiz-Villavicencio and Sant’Anna 2025](#)). At the same time, this was a substantive reform to the country’s bankruptcy regime, and so it is likely that the timing of the reform could have been anticipated. For these to affect causal identification, the anticipation must vary within firms by their type across districts with differing judicial capacities. I implement the research design as a dynamic event study where I find no evidence of such anticipation behavior. Furthermore, I also include district-year and industry-year fixed effects to account for any location or industry-specific dynamics that could be affected by the reform. Lastly, I examine the effects of the bankruptcy reform by district judicial capacity for the average firm (i.e. estimate difference in differences) to enable interpretation from the triple differences estimates in the absence of an “ideal” control group unaffected by the reform.

I use multiple datasets for my analysis. First is a firm-level annual balanced panel data from a representative sample of formal sector incumbent firms in the CMIE Prowessdx database. I use a firm’s credit rating prior to 2010 to measure default risk. Specifically, I classify a firm as high-risk if their credit rating is ever in one of the 3 risk categories defined by national credit rating agencies in India. I classify firms based on their capital efficiency using a two-step process. I first estimate their MRPK using annual sales, value of plants and machinery, and wage bill assuming Cobb-Douglas production ([Hsieh and Klenow 2009](#); [Bau and Matray 2023](#)). Next, I scale their sales to capital ratio by the estimated capital share to calculate MRPK. Lastly, I define high-MRPK firms as those with above median estimated MRPK. While lending based on credit rating is common, it isn’t obvious that the lenders optimize on capital efficiency. I use estimated MRPK as the planner’s benchmark for capital efficiency while emphasizing that a lender who maximizes repayment rather than social return has no private reason to allocate credit along this margin.⁴ This distinction is central to my interpretation: I test whether the reform along with local enforcement capacity reduced credit towards high-risk firms and/or moved credit toward high-MRPK to examine whether this affects credit disciplining or allocative efficiency or both.

The second dataset, and related to firm behavior, is bankruptcy filing data under IBC. I examine publicly available bankruptcy filings (claims) available with the Insolvency and

⁴MRPK is an imperfect proxy for capital efficiency: it equals the marginal product of capital only under a common Cobb-Douglas technology, is built from sales revenue rather than physical output, and uses book values of capital. My interpretation does not hinge on MRPK being measured precisely but rather I use it to generate ordinal ranking of firms by their sales-capital ratio. I examine robustness to alternative proxies in [Section 6.4](#).

Bankruptcy Board of India (IBBI), which I merge with firm characteristics in the Prowess sample using unique corporate registration identifiers in both datasets.

The final dataset is the district court-level data from [Ash et al. \(2025\)](#) and [Rao \(2025\)](#). I measure district-level judicial capacity as the predetermined judge to population ratio. This measure reflects the stark undersupply of judges to resolve legal cases, where the national average is 20 judges per million compared to over 100 in advanced economies like the United States. This is also a commonly used metric to measure judicial capacity, including in cross-country comparisons such as [Wittrup 2024](#). The low national-level average masks substantial variation between districts, where some districts have higher ratio than others. I match Prowess sample of firms with the district court data based on a firm’s predetermined registered office location to minimize concerns about endogenous sorting.

This paper presents three main findings. First, I find that borrowing decreases but net short-term liquidity increases among high-default-risk firms in better judicial capacity districts following the reform. On the other hand, I do not find any clear effects on borrowing or liquidity among high-MRPK firms in similar districts. I note that a firm’s classification of default risk is not strongly (negatively) correlated with its estimated MRPK even after accounting for industry and location of the firms. I rationalize these results on borrowing and liquidity by acknowledging that firms classified as high default risk could be different from low-MRPK firms. I interpret increased liquidity as suggesting that high-default-risk firms hold more cash at hand in districts with better enforcement after the reform. This result is consistent with the findings of [Vig 2013](#); [Closset et al. 2023](#), which documented that strengthening creditor rights increased liquidity hoarding among firms in a similar context with multiple market failures.

Second, I note significant contractionary effects on the economic outcomes of high-default risk firms, including their legal and interest expenditures, investment, and sales revenue. Using bankruptcy filing data, I find that high risk firms are more likely to be liquidated on average but less likely in better judicial capacity districts. Squaring this with the borrowing results, this could plausibly indicate better credit behavior in high capacity districts, where such firms use increased liquidity to repay past dues. I find no effects on the expenditures or sales revenue among high-MRPK firms and a positive effect on long-run investments. One plausible explanation for these results among high-MRPK firms is that their capital efficiency is unobservable to lenders, suggesting continuing information frictions. These firms are also not likely to file for bankruptcy, at least during the study period.⁵

⁵The firm-level investment variable in CMIE Prowess dataset includes investments in both productive

Third, I examine judicial outcomes of cases involving litigating firms including banks using case-firm merged data from [Rao \(2025\)](#). I show that district courts enable *execution* of existing judgment orders and complement specialized courts for debt recovery and bankruptcy even if they themselves don’t directly adjudicate these matters. Overall, I find increased resolution of firm-level cases in district courts with better judicial capacity, particularly in cases involving *high-default-risk firms as defendants*. I find suggestive evidence of increased resolution of bank-level cases when banks appear as plaintiff although this result is not significant at conventional levels.

These contrasting results - credit disciplining effects without the relative increase in borrowing by high MRPK firms - suggest that well-enforced bankruptcy reforms may not on their own redirect credit towards higher MRPK firms, consistent with binding information frictions. These findings are consistent with a simple model of credit allocation in the presence of varying institutional quality and incentives faced by lenders (bank or loan officials). Typically, bank officers have lower incentives to declare a default and often provide additional loans to refinance older, delinquent debt. This phenomenon is referred to as “ever-greening” and was prevalent pre-IBC in India ([Chopra et al. 2020](#); [Kulkarni et al. 2025](#)) and is a common global phenomenon ([Jordà et al. 2022](#); [Li and Ponticelli 2022](#); [Faria-e-Castro et al. 2024](#)). The bankruptcy and associated reforms (such as the AQR) along with local judicial capacity could reduce the pressure on ever-greening but it need not present mechanisms to reduce information constraints. Consequently, the reform may result in relative and not absolute improvement in allocative efficiency if lending decisions based on capital efficiency remain unchanged.

A natural concern is that the triple difference design identifies *relative* movements—how borrowing by high-default-risk firms shifts relative to low-risk firms in high- versus low-capacity districts—rather than changes relative to the first-best allocation. I address this in two ways. First, the triple difference identifies the *margin* along which credit is reallocated, focusing on separating an enforcement channel (reallocation on observable default risk) from an information channel (reallocation on unobservable productivity). Second, I recover the “missing intercept” (what happens to average borrowing) with a difference-in-differences on the average firm: total borrowing *rises* in high-capacity districts after the reform. This suggests that credit was differentially allocated *along the risk margin*, after the reform.

and financial projects. In the absence of borrowing effects, high MRPK firms could plausibly invest their retained earnings and cash from other reduced costs in long-run projects.

This paper contributes to two main strands of the academic literature. First, it contributes to the literature on credit misallocation by bringing attention to the tension between information and enforcement frictions (Karlan and Zinman 2009; Blouin and Macchiavello 2019). Regulatory state lending and bailout policies (Banerjee and Duflo 2014; Giné and Kanz 2018; Bau and Matray 2023), and incentives faced by banking officials (Cole et al. 2015; Fisman et al. 2017; Chopra et al. 2020; Faria-e-Castro et al. 2024; Naaraayanan and Wolfenzon 2024; Kulkarni et al. 2025), could focus more on enforcement relying on political or individual incentives (Khwaja and Mian 2008; Lilienfeld-Toal and Mookherjee 2016) and less on allocative efficiency (Müller and Verner 2023; Jack et al. 2023). This paper demonstrates that while a strong bankruptcy regime requires complementary investment in judicial capacity to reduce lending towards high-default-risk firms, it isn’t enough for increasing allocation towards more productive firms, particularly when productivity is potentially unobservable at the time of lending.

Second, I show that the last-mile enforcement capacity of *general-jurisdiction* district courts determines whether a national creditor-rights reform changes firm behavior (La Porta et al. 1998; Rajan and Zingales 1998; Djankov et al. 2007; Giné and Love 2010; Vig 2013; Gormley et al. 2018; Closset et al. 2023; Bhupatiraju et al. 2024). This builds on the well-documented result that better courts relax credit constraints (Visaria 2009; Ponticelli and Alencar 2016; Amirapu 2021; Müller 2022; Alok et al. 2022; Li and Ponticelli 2022; Rao 2025). Two features distinguish my result. First, the reform established dedicated bankruptcy tribunals (NCLT/DRT), yet its credit disciplining effect is concentrated in districts where ordinary local courts have greater capacity. This is because these courts execute recovery orders and resolve ancillary disputes (for example, dismissing challenges to liquidation or asset valuation orders) through which creditor rights are ultimately realized. Specialized adjudication does not substitute for local enforcement capacity. Second, this local capacity operates on a specific *credit-disciplining* margin, complementing the *credit-access* margin emphasized in prior work.

Rest of the paper is structured as follows. [Section 2](#) describes the banking sector, lender rights and enforcement mechanism in India under pre-reform and post-reform regimes. Empirical strategy is laid out in [Section 3](#). [Section 4](#) describes the datasets and variables used in the analysis. [Section 5](#), [Section 6](#), [Section 7](#) report results, robustness, and mechanism, respectively. [Section 8](#) concludes.

2 Context

The banking sector in India is characterized by a dominant share of public sector banks and a small share of private banks, together known as Scheduled Commercial Banks or SCBs, which are regulated by the central bank, Reserve Bank of India (RBI). The state has always played a key role in the evolution of this sector through varying systems of controls on the sector's operation including nationalization or the dominant public ownership and controls over lending operations through priority sector lending norms , liquidity and cash reserves requirement (Demetriades and Luintel 1996). Borrowers, particularly agricultural and small firms, are also frequently bailed-out by the government prior to elections, eroding timely repayment behavior (Giné and Kanz 2018).

Non-performing assets (NPAs) in India have historically been over 2% and steadily increasing since 2012, peaking to slightly above 11% (Figure 1). The peak roughly corresponds to the timing of bankruptcy reform and other regulatory changes intended to address this problem of growing NPA.

The bankruptcy process in India until 2016 favored promoters/founders and shareholder rights over lender rights during restructuring or liquidation proceedings. The 2016 Insolvency and Bankruptcy Code (IBC 2016) is a new consolidated law on bankruptcy process that provides for a market-based mechanism for time-bound resolution, emphasizing restructuring over liquidation, and at the same time prioritizing the rights of financial lenders such as banks in the recovery of unpaid debt from the borrower.

Figure 2 provides a timeline of the reform. Up until 2016, codes governing bankruptcy proceedings were fragmented across many statutes. Company Law, which lays down the rules governing incorporated entities, was amended in 2013 in an attempt to streamline the process. An all encompassing bankruptcy statute under IBC was tabled in the Parliament in the winter session of 2015, passed and became a law by May 2016. The first set of bankruptcy resolutions were passed in 2017. I use the year of the law (2016) to divide the time period into pre-reform period when the rights of financial lenders were weak and post-reform when the lenders were made key stakeholders in resolving debt defaults, particularly large scale corporate debt defaults, to decide on restructuring or liquidation as the appropriate next step.

There were many concurrent policies by the RBI, such as the 2015 Asset Quality Review and the 2018 guideline in promptly identifying defaults and initiating reporting towards bankruptcy processes. All of these policies aimed to make the IBC 2016 effective. Therefore,

the interpretation of a change in bankruptcy regime that I examine in this paper includes not just the on-paper reform but also all complementary policies aimed at improving the resolution of stressed capital.

2.1 Role of the Judiciary in Bankruptcy

The dispute resolution institutions relevant to the bankruptcy regime in India include general courts of law, specialized debt recovery tribunals (DRTs) and specific bankruptcy courts - National Company Law Tribunal (NCLT) and National Company Law Appellate Tribunal (NCLAT) - that were set up under IBC 2016. These together are responsible for implementing the law and for providing clarity over both the process and outcomes for borrowers and lenders. Frontline district courts are the relevant institutions in the recovery of smaller debts (under INR 2.5 million) and for the execution of judicial orders. These district courts are also ubiquitous - one for each administrative district - compared to only 26 unique DRTs locations across India or 16 locations of the NCLT. District courts make access to judicial institutions easier when defaulters are based in non-metropolitan districts far away from the location of DRTs or NCLT benches (see [Figure A.1](#) for the spatial distribution of district courts in this study sample and DRT/NCLT benches).

The role of general jurisdiction frontline district courts is particularly important as many aspects of the new bankruptcy regime are yet to be implemented in practice and to resolve concerns arising from the complex interaction between the new bankruptcy code and older laws, including procedural code (for example, the 1908 Code of Civil Procedure) or from specialized laws such as SARFAESI Act, 2002 ([Fig 2013](#)). In fact, around 44% of the liquidation orders remain stuck in various litigation in general jurisdiction district courts. Perhaps due to these challenges 58% of bankruptcy resolutions recommend restructuring relative to 42% favoring liquidations, reducing credible threat from implementable liquidation and relying on restructuring as second-best reform strategy ([Insolvency and Bankruptcy Board of India 2026](#)). As a result, these courts play a complementary role in the enforcement of lender rights even under the new bankruptcy regime and their capacity could matter for the subsequent impact of the reform.⁶

The court in which a suit should be filed is defined in the Code of Civil Procedure, 1908. Neither the lender nor the defaulting borrower has any choice over where they can file their

⁶For example, I note multiple cases pertaining to a specific bankruptcy law (from prior to the 2016 reform) in district courts using the [Ash et al. \(2025\)](#) dataset. These cases invoke Section 17A of the SARFAESI Act, 2002, which allows firms or individuals to challenge liquidation of their assets by financial institutions in their local court. Additionally, over a million legal cases in district courts pertain to execution of past judgements/orders.

lawsuit. The bankruptcy process itself is adjudicated by the NCLT (a centralized tribunal), where the suit can either be filed by the lender or initiated by *corporate* borrowers either through asset restructuring or liquidation as the last resort. However, bankruptcy proceedings related to individual borrowers under the Provincial Insolvency Act (1920), or liabilities from criminal offense, fraud and family disputes, personal insolvency, anti-trust and intellectual property, and property related disputes such as tenancy and eviction continue to be litigated within the district courts.

How well the district courts function depend on the staffing levels in such courts (Rao 2025). While there is an active debate and discussion on how best to arrive at the “optimal” number of judges per court (*Imtiyaz Ahmed v. State of U.P. , 2012*), the method of allocating number of judges per court during the study period followed a rule of thumb approach of 50 judges per million people (*All India Judges’ Association case, 2002*). In practice, many courts fall short of this number and there is wide variation in the realized number of judges per capita across districts in India (see Table 1 on summary statistics). A lot of this variation is due to decentralized approach to judicial administration including budgetary and implementation support that is provided by respective state governments. While states with higher tax-to-GDP ratios could have higher number of judges per million, competing policy priorities for funding other projects/programs, delays in recruitment, and other infrastructure bottlenecks (lack of adequate courtrooms or physical space) could drive the observed variation. I account for these state and district-level dynamic considerations in allocating judicial resources in my empirical design that I discuss in detail in Section 3.

2.2 Bank Officers’ Incentives

Credit allocation is likely driven by monitoring and career incentives of bank officials. Since employees of public sector banks are tenured officials, who are subject to frequent transfers to different branch locations, their incentives to sanction loans may not align with the objective of efficient credit allocation but rather to maximize their own personal incentives such as enforceability of repayment (Fisman et al. 2017). Furthermore, initiating bankruptcy process requires lenders to make provisions for any expected losses, which may be at odds with the incentives of the bank official that depends on the extent of surplus on the bank’s balance sheet (Kulkarni et al. 2025). Acknowledging that such incentives could limit the impact of the reforms, India’s central bank, the RBI, introduced the AQR and additional guidelines to all banks to tighten supervision and reporting of defaulted loans in a timely fashion, within a few months following the first set of bankruptcy resolution (Kulkarni et al. 2025). However, these guidelines did not change lenders’ incentives to lend more to entities with higher capital

efficiency and thus, the impact of the reform on the efficiency of credit allocation is not apriori clear.

3 Empirical Strategy

I follow a triple difference (DDD) research design to test the role of local judicial capacity with respect to the bankruptcy reform, which could affect credit allocation by firm type. I estimate the following specification using predetermined variation in firm default risk (based on predetermined credit rating) or capital efficiency (estimated MRPK) and variation in local judicial capacity measured as judge to population ratio.

$$Y_{fidt} = \beta_1 \text{Hi Risk}/\text{MRPK}_f \times \text{Hi-Jdg-Pop}_d \times \text{Post}_t + \beta_2 \text{Hi Risk}/\text{MRPK}_f \times \text{Post}_t + \delta_f + \delta_{it} + \delta_{dt} + \epsilon_{fidt} \quad (1)$$

Y_{fidt} represents relevant outcomes (borrowing or production) of firm f , in industry i , registered in district d in year t . Firms are classified as high-default-risk or high-MRPK as defined in [Section 4](#), where Hi Risk_f is a dummy variable taking value 1 if their credit rating falls in any of the 3 risk categories at least once before 2010. Similarly, Hi MRPK_f is a binary variable based on categorization of firms as above median in the distribution of pre-2010 estimated MRPK across the sample firms (coded as 1 and below median coded as 0). Districts are classified as high judicial capacity, which is denoted by dummy variable High Jdg Pop_d that takes value 1 if the judge-to-population ratio in the 2012-2016 period is above the sample average of 51.7 judges per million, and 0 otherwise. This threshold also maps to the recommended staffing levels by the Supreme Court of India. Post_t is a dummy indicating whether year $t > 2016$ (i.e. period after the reform), defined as $\text{Post}_t = 1$, for $t > 2016$ and 0 otherwise. Since the data follows a panel structure, I include firm fixed effects denoted by δ_f to account for all firm-level, time invariant unobserved potential confounders. In addition, I flexibly account for unobserved dynamic confounders at industry and local district levels using industry-year (measured as 2 digit NIC product code associated with each firm) and district-year interactions. These are denoted by terms δ_{it} and δ_{dt} , respectively. ϵ_{fidt} is the idiosyncratic error term. Since the “treatment” varies at the district level, I cluster standard errors by district. I also bootstrap standard errors when using estimated-MRPK-based design.

The main null hypothesis of interest is whether the coefficient $\beta_1 = 0$ in [Equation 1](#) above. A negative and significant coefficient would imply that the outcome declines in the post

reform period in districts with better judicial capacity for a firm belonging to the specific category relative to those not. Similarly, a positive coefficient would imply that the outcome increases in the post reform period in district with better judicial capacity for firms of a specific type relative to those not of that type. Credit disciplining (for example, based on result 1 from [Appendix A.1](#)) implies that $\beta_1 < 0$ for high-default-risk firms. Moreover, if we expect allocative efficiency to improve (which is not explicitly modeled), then we expect $\beta_1 > 0$ for high MRPK firms.

To interpret β_1 as a causal parameter, the assumption is that borrowing (and any other outcomes, Y_{fidt}) by hi-default-risk (high MRPK) trends in parallel to those by low risk firms (low MRPK) within high judicial capacity or low judicial capacity districts, even if, on average, they don't trend in parallel to each other. That is, the design relaxes the assumption that high-risk firms trend similarly to low-risk firms everywhere. For example, it is plausible that high-risk firms could experience more unobserved shocks over time due to their behavior, which could put them on a different trend than low-risk firms. While this would be problematic in a difference in differences (DiD) design, the DDD design allows such differential trends as long as the same trends are observed in high and low judicial capacity districts. Furthermore, I include industry-year and district-year fixed effects in order to control for any unobserved time-varying confounders by industry or geography, such as variation in resource allocation to district courts over time. Finally, because the bankruptcy reform was a one time, pan-India reform with no staggering of treatment, I do not impose additional restrictions of homogenous treatment effects across space and over time.

I implement [Equation 1](#) using two-way fixed effect specification (firm and year fixed effects) with industry-year and district-year interactions, defining post-period to include years following the reform in 2016. I also implement the design using an event study specification, presenting β_1 for each year-bin prior to and post reform. The event study implementation allows examining the dynamic effects of the reform across varying judicial capacity by firm type and also test for the parallel trends assumption within district groups by judicial capacity.

Interpreting the estimand: The coefficient β_1 identifies a *relative* object: the change in the high-minus-low-type outcome gap, before versus after the reform, in high- relative to low-capacity districts. Two implications follow. First, the design differences out any district-wide or economy-wide time-varying changes in credit and other unobserved confounders, so β_1 is specific to the relative effect. Furthermore, the average effect of the reform is absorbed by (i.e, multi-collinear with) the district-year fixed effect. β_1 does not provide information

relative to a first-best level but rather relative to low risk or low MRPK categories. Thus, this design does not tell us whether high-default-risk/MRPK firms were over- or under-borrowing before the reform. What β_1 identifies is the margin along which enforcement-backed creditor rights reallocate credit, holding fixed everything common to a district-year. Second, I recover the enforcement effect of the reform for the average firm, one that is not estimated by Equation 1 using a difference-in-differences (DiD) design comparing high- and low-capacity districts before and after the reform. The DiD shows the net effect of the reform enforcement on credit - whether increased, decreased, or remained the same - in order to interpret β_1 appropriately.

3.1 Testing Identifying Assumptions for Causal Inference

Since the reform was a one-time, pan-India reform, all districts and firms experienced the regime change at the same time. This allays any concern arising from staggered roll-out research designs that require strong assumptions for causal identification. However, it is plausible that local actors such as banks could lobby to improve their respective courts' judicial capacity by asking for more judges in areas with more efficient or high risk firms in order to initiate bankruptcy processes. This would make judicial capacity endogenous to the bankruptcy regime. I address this concern by using pre-reform judge to population ratio as a pre-determined measure of local institutional capacity to test my main hypotheses.

I test for the assumptions for causal inference in two ways. First, I examine whether there are any differential trends in outcomes between high risk/MRPK firms and low risk/MRPK firms across districts with varying judicial capacities before the reform (i.e., look for any “pre-trends” in the pre-reform period in the DDD set up). Second, I estimate and examine the prior-period trends in a simple DiD framework by examining outcomes across all firms between high and low judicial capacity districts before the reform. This is to test whether the composition of firms and their dynamics prior to reform could confound the post reform effects in the DDD specification.

Another concern is contemporaneous expansion of bankruptcy courts (NCLT and its regional benches) following the 2016 reform, which could confound whether the results are driven by local district court capacity. This concern is effectively addressed due to the specific context of this study as the newly introduced bankruptcy courts had a more aggregate jurisdiction than that of a district court (it was even larger than a state). At the time of the reform in 2016, the federal government introduced 10 regional benches and one principal bench (New Delhi), which was slowly expanded to 16 benches in total as of 2026. Since the geographic

scale of these benches is much larger than a district, with these specialized courts mainly located in state capitals or large urban centers, the empirical specification [Equation 1](#) exactly accounts for such concerns through the district-year fixed effects. Consequently, the observed estimates are to be inferred as additional effects of local court capacity over and above any simultaneous effects of new bankruptcy courts.

Furthermore, inclusion of district-year fixed effects when the outcome is measured at a firm-year level also addresses concerns about potential, time-varying unobserved confounders like district wealth levels or urbanization trends in driving better judicial capacity over time. The main assumption for causal identification, therefore, is that there are no *differential* trends in how economic agents like banks respond to firms by their default risk or capital productivity in districts with better or worse judicial capacity independent of the bankruptcy reform or its enforcement. I show the validity of this assumption through multiple falsification tests that I discuss later in [Section 6](#).

3.2 Effect on Litigation Outcomes

Since one of the key focus in this paper is how institutional quality matters in credit allocation, I use legal case-level data, merged with firms' identity to examine how local courts resolve cases pertaining to firms of specific type (high-default-risk or high-MRPK) or those pertaining lenders (banks) before and after the reform. As the case records allow me to identify the litigant type of firm involved in the case - as plaintiff or defendant, I estimate the triple difference specification [Equation 1](#) to examine firm-specific litigation outcomes - number of cases filed, resolved, or pending - separately by litigant-type.

For banks, I estimate a DiD specification instead of a DDD to examine bank-specific litigation outcomes, separated by litigant-type (for cases where banks appear as a plaintiff or as defendant), to examine how debt recovery cases are handled by courts of differing capacities before and after the bankruptcy reform. This slightly different approach is required to examine lender's (i.e bank) recovery behavior through courts. Since banks operate through their location-specific branch offices to make lending decisions, enforcement quality by local courts could matter even for national commercial banks.

4 Data

I use multiple datasets. First, I use [Ash et al. \(2025\)](#) and [Rao \(2025\)](#) datasets to measure judicial capacity and judicial outcomes, respectively. Second, I use CMIE Prowess dx dataset to measure firm-level characteristics to study allocation and primary outcomes of interest -

borrowing, short-run liquidity, and production outcomes. I merge the two datasets by the district identifier of firms’ registered office. The third dataset includes bankruptcy filing data under the new IBC, 2016 law. I describe these data and variables constructed for analysis in greater detail below.

4.1 Judicial Capacity and Judicial Outcomes

Judge to population ratio is a standard measure of judicial capacity used in the literature (see [Wittrup 2024](#); [India Justice Report 2025](#)). More broadly, staff-per-capita is a commonly used metric to reflect underlying state capacity in the absence of data on service quality or efficiency metrics. For example, [Chalfin and McCrary \(2018\)](#) show that number of police officers per capita is a strong determinant of crime reduction in large cities in the United States. [Needleman et al. \(2002\)](#) discuss how shortage of nurses affects health outcomes and mortality of patients. In the context of India, [Dhaliwal and Hanna \(2017\)](#) show that increased attendance of nurses in primary health centers reduced children born with low birth weight. More directly related to the context of this paper, [Rao \(2025\)](#) shows that backlog in district courts reduces when more judges are added to a court.

I use the universe of legal case data from [Ash et al. \(2025\)](#) to calculate district-level judicial capacity measured as judge-to-population ratio. I use the judge data table to compute total number of judges observed prior to the reform using judge tenure start and end dates. The data table includes the following variables: state, district, court establishment (or simply the court), and judge position. For each judge position within a court, the data also contain the start date and the end date of the judge assigned to the position.⁷ Next, I create a dummy variable assigning a value 1 for every calendar year between the start and end years for the assigned judge, and 0 for calendar years outside this range. I do this for all of state-district-court-judge position-start year-end year tuples over 2012-2016 to calculate the maximum active judge positions in the respective district and normalize relative to the 2011 district population. Finally, I define districts with above average value (51.7 judges per million) as “high judge population ratio” districts. This also maps to districts that are above the recommended staffing levels by the Supreme Court of India.⁸

To measure litigation-specific outcomes such as filing rates, resolution rates, and pending cases at firm-level, I use data from [Rao \(2025\)](#) containing all meta-data and identifying

⁷The public version of the data does not contain any identifiers such as judge or litigant names.

⁸The average judge to population ratio based on this approach is 18.6 judges per million using all districts, which is very close to the aggregate number of 20 reported by the Supreme Court of India. The average ratio for districts in the study sample, which correspond to the matched firm-location is higher at 51.7.

information from the universe of legal case records from a sub-sample of 195 districts over the same time period. I match firm-level identifiers with the legal case-level data from this dataset using firm name and place of registration, followed by manual verification of the resulting match. I identify over 3500 unique non-financial firms with cases across these courts with different baseline levels of judge-to-population ratios. The metadata allows me to tag each firm-case record to identify whether the firm appears as a plaintiff or as a defendant. For each combination of firm-court-litigant type, I calculate number of new cases filed, number of cases resolved, and total pending backlog using filing and decision time stamps in the meta-data. I follow similar algorithm for cases involving banks and financial institutions.

4.2 Firm-level Data

I use CMIE Prowess dx dataset that contains a representative sample panel data of corporate entities (registered formal sector firms) in India with annual financial and production data since 1990s. This dataset offers the advantage of a large-scale firm-level panel data for India, which includes company name, registration details in addition to annual financial and production information. It represents multiple sectors including manufacturing, trade, and services, and thus suitable for research questions requiring firm-level information and their fine-geographic footprint. I am interested in non-financial sector firms, which form the main sampling frame for our analysis. I merge these firms with the district-level judicial data by firms' registered office location to generate the analysis sample.⁹

For primary outcomes, I focus on firms that were incorporated prior to 2010 and continue to exist through the study period. This allows me to construct a balanced panel of incumbent firms with annual outcomes spanning 2010-2024, while minimizing concerns about compositional changes to the sample of firms over time. The main analysis sample contains a balanced panel of 3500 firms located across 243 districts in India. To address concerns about composition of firms in the sample, particularly due to firm exits, I also examine survivorship of firms following the reform by underlying judicial capacity.

In order to examine credit allocation across the sample firms, I classify firms based on their default risk and capital productivity from credit ratings and estimated MRPK using data prior to 2010, respectively. Using pre-2010 data introduces a gap of at least 5 years between measurement and the reform, which could potentially address anticipation concerns in the

⁹Other firm-related datasets from India are the Annual Survey of Industry (ASI) and the Micro-Small and Medium Enterprise (MSME) datasets, neither of which provide firm-level identity and their specific geographic location. However, the key trade-off in using Prowess is that it excludes smaller firms and those in the informal sector. Information from these can mainly be collected through primary enterprise surveys, often requiring large funding for data collection.

context of a large-scale, national-level reform.

To measure default risk, I use firms’ historic credit rating information. Each year, firms are rated based on their past performance on credit contracts by credit bureaus. There are five credit bureaus - CRISIL, ICRA, CARE, IND-RA, and BRICKWORK. These are similar to global agencies like Moody’s and Standard & Poor’s. Ratings like {PR 1+, A1+, A1, PR2, A+, A, A-, ... }, which map to specific credit safety measures are classified into 8 bins in decreasing order as “Highest Safety, High Safety, Moderate Safety, Adequate Safety, Inadequate Safety, High Risk, Substantial Risk, and Default”. I classify firms as high-default-risk if their credit rating in the pre-reform window 2005-2010 is marked “High Risk” or worse (i.e., High Risk, Substantial Risk, or Default).

I estimate firm-level MRPK by assuming Cobb-Douglas revenue function as in the literature (for example, [Bau and Matray 2023](#)) to estimate input elasticities and then construct the marginal revenue product of capital. This is a two-step process where in the first step, I estimate the input shares in the revenue function using panel regression of log sales on log capital (value of plants and machinery) and log labor (wagebill) with firm and year fixed effects. In the second step, I take the estimated coefficients on capital as the common output elasticity ($\hat{\alpha}$) and compute firm-year average revenue product, $(\frac{Y}{K})$. Finally, I define $MRPK_{it} = \hat{\alpha} \cdot (\frac{Y_{it}}{K_{it}})$. This yields firm-level measures of MRPK used for classification and heterogeneity analysis. Since this estimated measure is a continuous variable, I classify firms as high-MRPK if their estimated MRPK in the pre-reform window 2005-2010 is above the median value across all sample firms. Panel A [Figure A.10](#) shows similar distribution of estimated MRPK across all sample firms and firms in the manufacturing sector where capital is more reliably measured.

Bankruptcy Filing Data I use publicly available bankruptcy filing data from IBBI website, which has details on the corporate entity under bankruptcy, including their unique corporate identifier or CIN number that can be used to merge with other sources of corporate data including Prowess. This data is only available following the IBC reform starting in 2017 since the filings were under the new law.

4.3 Outcome Variables

The first set of primary outcomes include borrowing and short-run liquidity (net working capital) to test credit allocation. The second set of outcomes include economic outcomes measured as legal and interest expenditures, investments, and sales revenue to examine pro-

ductivity implications of the reform and judicial capacity. I impute any missing values of the outcome measures using the firm-specific average during the study period.¹⁰

Since I do not observe actual firm exits in Prowess data, I use non-reporting or disappearance due the construction of the balance panel as a proxy to examine whether the reform differentially changes the composition of firms by underlying district judicial capacity. Similarly, I also examine “balance” using selection into reporting to ensure that the results are not driven by compositional changes.

Finally, I examine whether the sample firms are found in bankruptcy filings under IBC and whether the filing is under restructuring or liquidation using the universe of IBBI bankruptcy claims data. While this data need not provide a complete picture of firm exit since many of the proceedings are still ongoing, it provides an approximate measure of firm survivorship.

4.4 Summary Statistics

[Table 1](#) presents a summary of the court-firm matched dataset used in the analysis. The average judge to population ratio in the sample is around 52 judges per million. This reflects the spatial distribution of formal sector firms in India rather than the population of all district courts in India. Second, the sample firms are large, formal sector firms, where 80% of the sample are publicly listed on various stock exchanges. These firms exhibit a large variation in their default risk (average rating close to 6) and capital efficiency estimated as MRPK. Most firms are between 20-60 years old, 60% engaged in manufacturing and the remaining in trade and construction. Many of the firms in the Prowess dataset frequently are involved in litigation in district courts (see [Figure A.2](#)). [Table A.1](#) compares firms by default risk and capital efficiency using pre-2010 data across districts with high and low judge to population ratios. As expected, there is sorting of firms spatially, with better firms located in high judge-population districts on average. However, there are limited observed differences between high-default-risk firms across districts with differing judicial capacities.

¹⁰This approach is preferred over other methods such as industry-level imputation to preserve firm-level heterogeneity that is central to causal identification in this paper as also followed in [Link et al. 2024](#) for corporate investment variables. To ensure robustness, I examine selection into reporting in the data.

5 Results

5.1 Borrowing and Liquidity

I start my analysis with borrowing and short-run liquidity outcomes to examine firms' response to the change in bankruptcy regime and local enforcement capacity through district courts. Since default risk is typically observable as corporate credit ratings, I first estimate Equation 1 using default-risk classification. Panel A Figure 3 (left) presents the event study estimates and Col 1 Table 2 presents the DDD estimates. I find a reduction in borrowing by high risk firms in better judicial capacity districts, which persists in the long run (about 30% reduction). The post-reform average treatment effects on the treated (ATT) is about -16% (Col 1 Table 2).

An improvement in allocative efficiency from a planner's perspective would imply that more productive firms should borrow more if they are credit constrained, holding all else constant. Using estimated pre-reform MRPK, I am unable to reject the null hypothesis of no effect on relative borrowing by high MRPK firms in better judicial capacity districts after the reform. Both Panel A Figure 3 (right) and Col 3 Table 2 suggest muted effects on borrowing by high-MRPK firms relative to low-MRPK firms. This could suggest two alternatives: (a) these firms are not credit constrained and thus don't need to borrow, or (b) they are credit constrained but because their productivity is unobserved by lenders, more credit does not flow to them.

These relative estimates from DDD need to be read with the average-firm difference-in-differences to infer what happened to the overall credit market. Because total borrowing rises in high-capacity districts after the reform (Figure A.3), the negative high-default-risk coefficient in Col 1 Table 2 reflects a *recomposition*: credit released by high-default-risk firms is absorbed elsewhere in the same districts but not by high MRPK firms (Col 3 Table 2). It also speaks to the alternative that high-default-risk firms were borrowing *too little* to begin with: had the reform relaxed a constraint holding back good-but-risky projects, borrowing by these firms should have *risen* where enforcement improved; instead it falls. I therefore read the effect as credit *disciplining* response while remaining agnostic about whether the pre-reform allocation was efficient.

In contrast to the negative effects on borrowing, I find that high-default risk firms increase their short-term liquidity measured as net working capital. Panel B Figure 3 and Col 2, Col 4 Table 2 present the event study and DDD estimates. I find a persistent increase in short term liquidity by over 80% among such firms in better judicial capacity districts post

reform. Not only do high-risk firms borrow less, they are more likely to hold cash at hand as also documented as a consequence of improved creditor rights in the literature (Vig 2013; Closset et al. 2023). On the other hand, I find muted and statistically insignificant effects among high-MRPK firms in better judicial capacity districts (Col 4 Table 2).

I interpret these results within a context of limited overlap between the sets of high default-risk and low MRPK firms. I find weak negative correlation between a firm’s classification as high-default-risk and its classification as high-MRPK. Without any fixed effects, a high-default-risk firm is 9% less likely to be classified as a high-MRPK firm (Col 1 Table 3). After adjusting for industry or location or both, this correlation coefficient halves and loses statistical significance. As a result, any information on default risk, such as that provided by credit rating, is less informative about a firm’s capital efficiency or MRPK. This would be surprising in a world with perfect information where we would expect that high-MRPK firms should be more profitable and thus, less likely to default. However in a context with multiple factor market failures, this may not be surprising. For example, consider firms that have made huge, upfront investment and that expected future productivity includes some risk. Such firms are susceptible to business cycle shocks, which could lead them to default given their relatively weak cash position. Another exactly opposite example would be incumbent firms with moderate or even low MRPK but with consistent cash flow such that they are less likely to default. Both these examples illustrate hidden type and hidden action mechanisms that are not currently accounted in the computation of credit ratings, which is mainly based on observable and backward-looking measures such as leverage and past payment history. The positive liquidity effect is a plausible strategy to pay back any outstanding dues to avoid initiation of bankruptcy proceedings.

Thus, while better courts could lead to some improvement in credit allocation post reform - borrowing by high-default-risk firms decline relatively, indicating a plausible credit disciplining effect, the reform also falls short of reallocating credit based on capital efficiency.

5.2 Economic Outcomes

Figure 4 and Table 4 presents the estimates on economic outcomes from the DDD design both as event study and two-way fixed effects specification, respectively, comparing firms’ by their baseline default-risk across districts with varying judicial capacities. I find significant decline in such firms’ legal (17%, $p < 0.01$) and interest expenditures (33%, $p < 0.01$) in districts with better judicial capacity. I also find reduced investment by such firms, at least in the initial years following the reform, and a negative but noisy effects on their sales revenue.

Figure 5 and Table 5 presents the results using firms’s baseline MRPK across districts with better or worse judicial capacities. Consistent with credit outcomes, I find insignificant effects on legal or interest expenditures or on sales revenue. On the other hand, I find a significant increase in investment by such firms after the reform in better capacity districts. The investment effect could be driven by retained earnings by these firms or through other sources of financing such as trade credit in the absence of borrowing effects. This is plausibly consistent with *persistent information frictions*, where high-MRPK firms that are better at making long-run investments are essentially unobservable to lenders for an impact on their borrowing outcomes.

5.3 Firm Survivorship

Since the main analysis uses balanced panel of incumbent firms that survive the sample period, it does not address whether the bankruptcy reform and better local judicial capacity increased firm exits through liquidations.¹¹ Firm exits could affect how the sample firms respond to the underlying business environment beyond factors accounted for in district-year fixed effects and thus, could affect the interpretation of the estimates. I address this through two complementary approaches. First, I examine the universe of bankruptcy filings from the Insolvency and Bankruptcy Board of India (IBBI). This includes a larger set of firms including those in Prowess. With the caveat the sampling frames of bankruptcy filings and Prowess do not perfectly align and that the filing data are only available post reform, I merge the filing data with firm identifiers in Prowess using common corporate registration identifiers (CIN) available in both datasets. This merge allows me to examine whether a high risk firm registered in high judicial capacity district is more likely to file for bankruptcy. Table 6 presents the results. Consistent with credit disciplining effect, high default risk firms are more likely to file for bankruptcy and more likely to undergo liquidation after the reform on average. However, the filings and liquidation are *lower* in better judicial capacity districts. These results are suggestive of potential deterrence from the threat of well-implemented bankruptcy in such districts resulting in lower borrowing and higher liquidity among corresponding high default risk firms. I find no significant bankruptcy filing behavior among high MRPK firms.

Second, I use absence of annual reporting in Prowess data as a proxy for exit if firms have temporarily shut down but not exited. Figure A.4 and Table A.2 present the patterns in non-

¹¹If a firm is restructured, they continue to exist but with different structural adjustments to their financial and business management.

reporting among all Prowess firms as well as by their default risk/MRPK. The results suggest that if anything high risk firms are more likely report their financial data (or fewer missing observations) in high judicial capacity districts post reform. These positive reporting effects could also plausibly indicate asset restructuring instead of complete exit. I find similar lack of effects when I examine patterns in selection into reporting ([Figure A.5](#) and [Table A.3](#)).

6 Robustness of the Results

6.1 Role of Bank-Firm Relationship

The dataset used in the main analysis is firm-level and therefore, credit allocation behavior is inferred from observed borrowing and liquidity by sample firms over time. To examine bank-side action on allocation, I make use of firm-bank relationship reported in the “bankers” table in Prowess, which reports all banks, including commercial banks, regional banks, and credit unions, which a firm has had a banking relationship with. I generate two variables to examine the allocation patterns: First, whether there exists a relationship between a firm and a bank and second, total number of banking relationships by a firm. [Figure A.6](#) and [Table A.4](#) present the results from analyzing this relationship as a proxy for bank-side allocation decision using the triple difference specification in [Equation 1](#) ([Figure A.6](#) and Col 3-6 [Table A.4](#)). Further, because I have the identity of the banking partner for two specifications, I also include bank fixed effects as indicated in the columns.

This exercise indicates that there are no “extensive” margin changes in banking relationships by either high-default-risk firms or high-MRPK firms. The results on reduced borrowing amounts suggests that the relevant margin of action is the intensive margin, where banks may lend lower amount of capital to high-default-risk firms even if they do not eliminate their relationship with such firms entirely. Again, this is consistent with the focus of the bankruptcy reform on restructuring rather than complete liquidation.

6.2 Concurrent Shocks

In addition to the 2016 IBC code, there were many other concurrent policy shocks (other than those complementary to IBC such as the AQR or the 2018 circular) that could have differentially affected firms by their default risk or MRPK across high or low judicial capacity districts. Two significant shocks were demonetization launched in November 2016 and the roll out of new value-added tax regime (GST) in early 2017. Demonetization created cash shortages everywhere and it is plausible that high risk firms could have been particularly affected. To threaten identification, the shock needs to differentially affect high-risk/high-

MRPK firms in high judicial capacity districts. I address this in two ways: First, I exclude 2016-2017 time period from my event window since demonetization and GST roll-out were temporary shocks, and find that the borrowing and liquidity effects by firm type persist in the long run (see [Figure A.7](#)). Second, since these shocks were felt more at an industry-level where all firms within specific industries were more affected than those in other industries, I account for these industry-specific dynamic trends through industry-year fixed effects.

Banks differentially recognizing defaulting firms and filing for bankruptcy is the exact intention behind IBC. As a result, concurrent policies by RBI such as the AQR or the 2018 guideline are not specific threats to the empirical design but rather, serve as mechanisms behind the observed effect.

6.3 Other Threats to Identification

In all main event study implementation of the triple differences research design, the pre-period trends are flat. I also test for pre-trends in the primary outcomes among all-firms using a DiD design. I consistently note lack of pre-trends on borrowing as well as economic outcomes among all firms prior to the reform ([Figure A.3](#), [Figure A.8](#)).

I carry out additional tests using inframarginal firms as placebo in the triple difference specification, where firms are defined as inframarginal if their pre-reform credit rating is perfect. Borrowing or liquidity holding by such firms should not be affected by local institutional (district court) capacity and/or changes in lenders' incentives since these firms never default. [Figure A.9](#) presents the results from this analysis. I am unable to reject the null on borrowing or liquidity effects among these firms by district judicial capacity before and after the reform, suggesting that credit allocation likely changed for mainly for high default risk firms.

6.4 MRPK Estimation

There are two distinct concerns with respect to measuring a firm's capital efficiency: (a) classification of firms by their estimated MRPK is identical to classification based on sales to capital ratio because of Cobb Douglas production function and a single estimated capital share $\hat{\alpha}$, and (b) since the classification is based on estimated or generated regressor, this could import first-stage sampling error into the triple-difference standard errors, attenuating the treatment effects towards zero ([Pagan 1984](#)). I address these in multiple ways. First, the treatment in [Equation 1](#) is a *binary* indicator for above-median pre-reform MRPK, not

continuous MRPK. Because $\hat{\alpha}$ is a single output elasticity common to all firms and enters as $MRPK_f = \hat{\alpha} \cdot (Y_f/K_f)$, it is a strictly positive scalar that preserves the ranking of firms: the median split on $\hat{\alpha} \cdot (Y/K)$ is identical to the split on Y/K for any $\hat{\alpha} > 0$. First-stage uncertainty in the estimated MRPK is likely reduced in the binary indicator if the rank order of firms by their true capital efficiency is preserved. Estimating the specification either using classification based on MRPK or sales to capital (Y/K) ratio yields identical results. Without the functional form assumption, this translates to a classification based on firms' average revenue product of capital (ARPK).

Second, the results are also robust to relaxing the assumption of a single $\hat{\alpha}$ and using industry-specific $\hat{\alpha}$ as shown in [Table A.6](#). Furthermore, the results are robust to classification into high vs low MRPK within industry or pooled across industries when also using industry-specific $\hat{\alpha}$. I show that the kernel density distribution of estimated MRPK for all firms across sectors as well as those in manufacturing - where capital stock is better measured, share the same support. The adjusted and within- R^2 of 0.93 and 0.54, respectively, suggests that the production function explains more than half of the within-firm variation in sales revenue (see Panel A [Figure A.10](#)). When I estimate [Equation 1](#) on a subsample of manufacturing sector firms, where capital measurement is relatively more reliable, I find similar patterns: the reform does not appear to increase borrowing and liquidity among high-MRPK manufacturing firms in high judicial capacity districts (Panel B).¹²

Third, to address concerns with analytical standard errors, I recompute them with a two-step cluster bootstrap (districts as clusters, 1000 replications) that re-estimates $\hat{\alpha}$, re-derives the MRPK classification, and re-runs the triple difference in each replication. I continue to fail to reject the null on borrowing by high productivity firms in better capacity districts post reform.

6.5 Alternate Definitions of Judicial Capacity

These results are not driven by how I define judicial capacity. I find similar results if I use median value of pre-reform (2012-2016) judge to population ratio to classify as high or low judicial capacity instead of the average (see [Table A.7](#), [Table A.8](#), [Table A.9](#)). The choice of using the average is motivated by the Supreme Court of India's recommendation on judge-to-population ratio of at least 50 judges per million. I also use pre-reform vacancy rates among district judge positions as another way to measure judicial capacity. I find qualitatively

¹²There is an immediate dip in borrowing observed for both placebo and high-MRPK manufacturing firms in [Figure A.9](#) and Panel B [Figure A.10](#) respectively. These correspond to years with cash shortages including 2016 and 2017, which revert to the mean of no differential effect in the long run.

similar effects of the bankruptcy reform in high judicial capacity districts (defined as below median vacancy rates in the pre-reform period) on firm-level outcomes including borrowing, liquidity, and production outcomes (see [Table A.10](#), [Table A.11](#), and [Table A.12](#)). Lastly, it is plausible that judicial capacity measured prior to the reform could itself change over time if capacity constraints in other districts are eventually resolved. These district-level dynamics are accounted in district-year fixed effects, which reduces concerns about measurement errors arising from changing district judicial capacities over time.

7 Mechanism: Last Mile Enforcement by District Courts

A reasonable objection to the role of local district courts is that large, formal-sector firms of the kind in Prowess resolve debt recovery at specialized DRTs (established 1993) and, after 2016, the NCLT-rather than at general-jurisdiction district courts, so that district-court capacity should be irrelevant to them. My claim is not that district courts are the primary forum for adjudicating large-firm recovery or insolvency. Indeed, initiation and adjudication sit with DRTs and the NCLT. Rather, district courts perform a distinct and complementary function in the *last mile* of enforcement that these fora do not: they execute recovery certificates and decrees, adjudicate possession of and disputes over collateral, hear challenges and injunctions arising under the Code of Civil Procedure and Section 17A of the SARFAESI Act, and dispose of the ancillary litigation through which a paper order becomes an actual recovery. A firm can therefore be a heavy user of DRTs *and* still depend on its local district court’s capacity for that order to be realized. Consistent with this, [Figure A.12](#) shows that the district-court cases involving Prowess firms are concentrated in execution and collateral/possession matters rather than unrelated civil or criminal suits.

Using case-firm matched data from [Rao \(2025\)](#), I estimate the main identifying specification [Equation 1](#) for three main judicial outcomes - number of cases filed in the district courts, number of cases resolved, and number of pending cases - as the dependent variables, separately when the sample firm appears as a plaintiff and a defendant. Panel A [Figure 6](#) presents the results for high-default-risk plaintiff firms and Panel B [Figure 6](#) presents the results for corresponding high-risk defendant firms. Though not significant at conventional levels, these figures suggest that more cases are filed against high default-risk firms (i.e. as defendants) and there is increased resolution of cases against such firms in courts with better judicial capacity. There is no conclusive evidence on judicial outcomes for cases involving high-default-risk firms as plaintiff.¹³

¹³[Figure A.11](#) presents the results for plaintiff and defendant high-MRPK firms litigating in the sample courts, suggesting higher rates of resolution and lower filing rates among high-MRPK plaintiff firms.

Lastly, I examine legal cases involving banks, separated by banks as plaintiff or as defendant. I find that better courts are less likely to resolve cases against banks (when banks are defendants, see [Table A.15](#)). While the results are noisy, the coefficients when banks appear as plaintiffs suggest increased resolution of cases per bank.

These results suggest that district courts with better capacity are more likely to resolve cases with high-default-risk defendants, particularly in the last mile enforcement cases filed in local courts. In contrast, when such firms file counter-suits as plaintiff (for example, to contest a liquidation or an asset valuation order), better capacity courts do not appear to take any action.

8 Conclusion

This paper provides evidence in support of the complementary role local courts play in enforcing creditor rights in the context of a national-level bankruptcy reform in India. Using a triple-difference research design, I show that this complementarity brings in credit disciplining. I find that these results are suggestively explained by local courts supporting the bankruptcy process through the *last-mile* enforcement of judicial orders against debt defaulters.

The results on credit allocation likely imply continuing information frictions when capital efficiency is unobserved. This is evident in the weak correlation between observed firm-level default risk (credit ratings) and capital efficiency proxied by estimated MRPK. Moreover, lenders may care about debt repayment rather than optimize lending based on capital efficiency as per planner’s benchmark. Thus, well-implemented institutional reforms like the IBC in India in better judicial capacity districts could help recover dues from high default risk borrowers and reduce their relative borrowings, indicative of credit disciplining. At the same time, high productivity firms do not borrow relatively more than low productivity firms even though borrowing by the average firm increases in such districts.

As the law takes full effect over time, the long run and general equilibrium effects may be different. Furthermore, what this paper examines is capacity measured in terms of judge-to-population ratios, which is one specific aspect of judicial capacity. Other investments in the local enforcement environment, such as better triage through alternate dispute resolution and conciliation for enforcing any liquidation or asset transfers may have different impact, which remain as open questions for future research.

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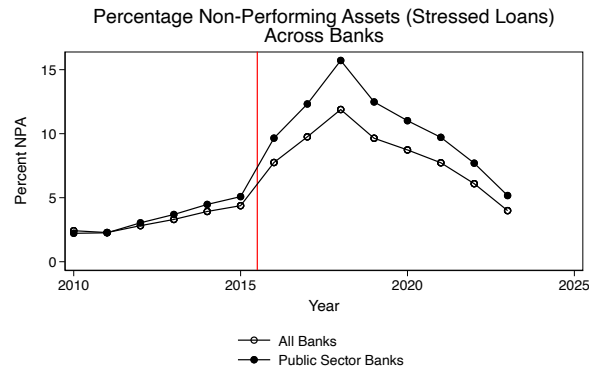
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9 Figures

Figure 1: NPA Across Banks in India



Notes: The figure above graphs the percentage of total lending declared as a non-performing asset (NPA) aggregated across all commercial banks in India and by public sector banks, as reported by the central bank, Reserve Bank of India. The red vertical line indicates when the bankruptcy bill was first tabled in the national parliament. I present the timeline of the reform below.

Figure 2: Timeline of Bankruptcy Reform

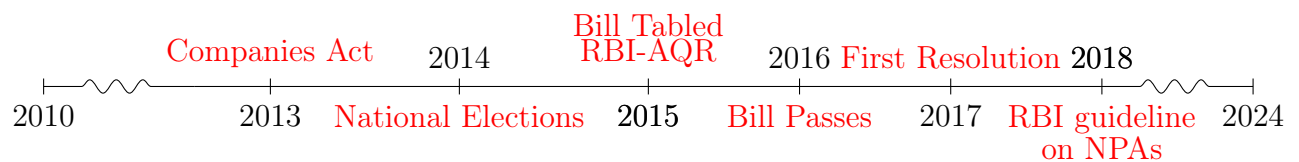
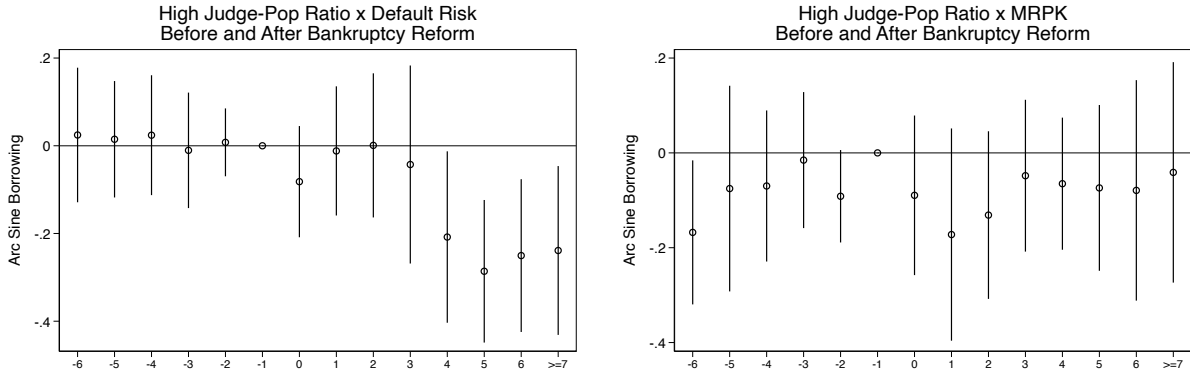
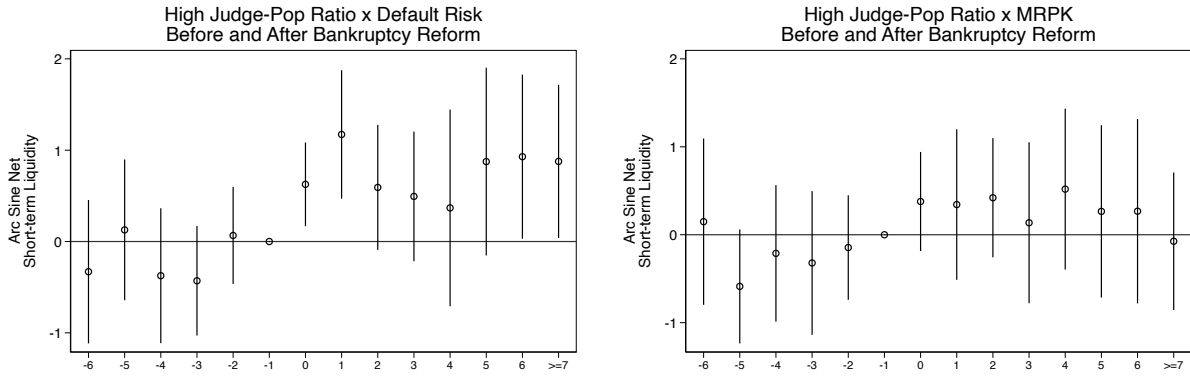


Figure 3: Borrowing and Liquidity Outcomes

Panel A: Borrowing

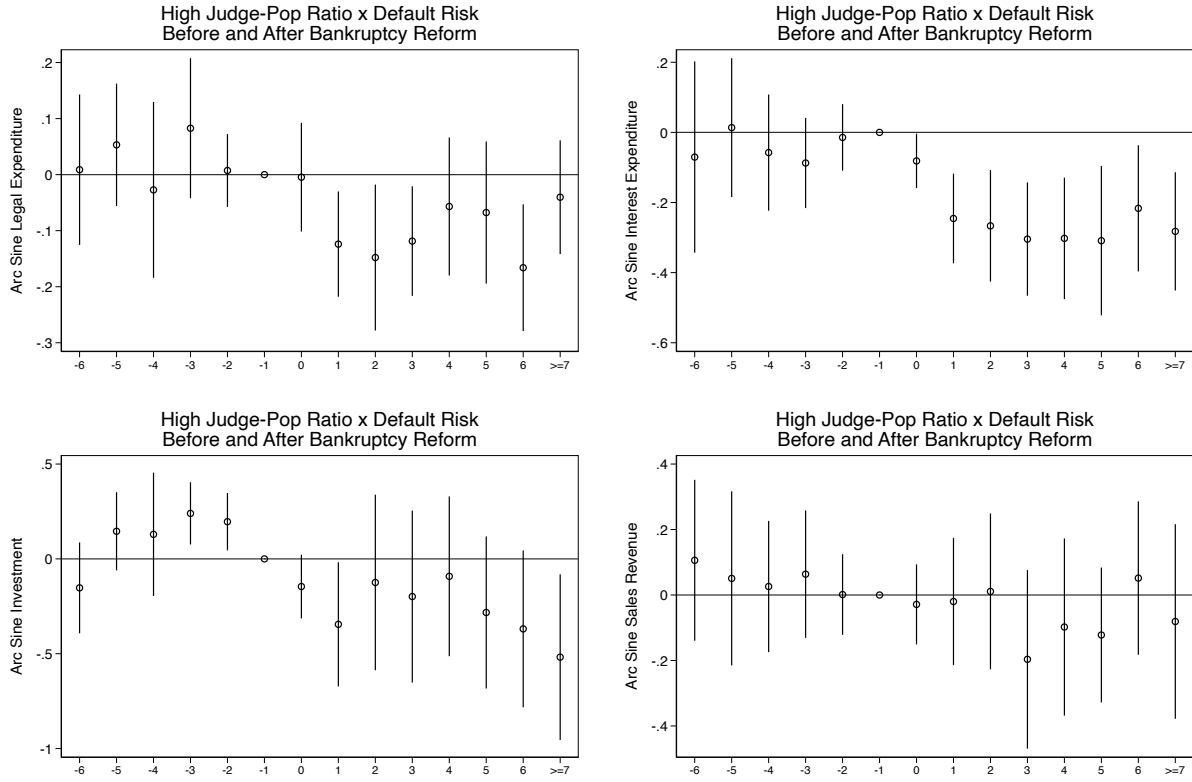


Panel B: Liquidity



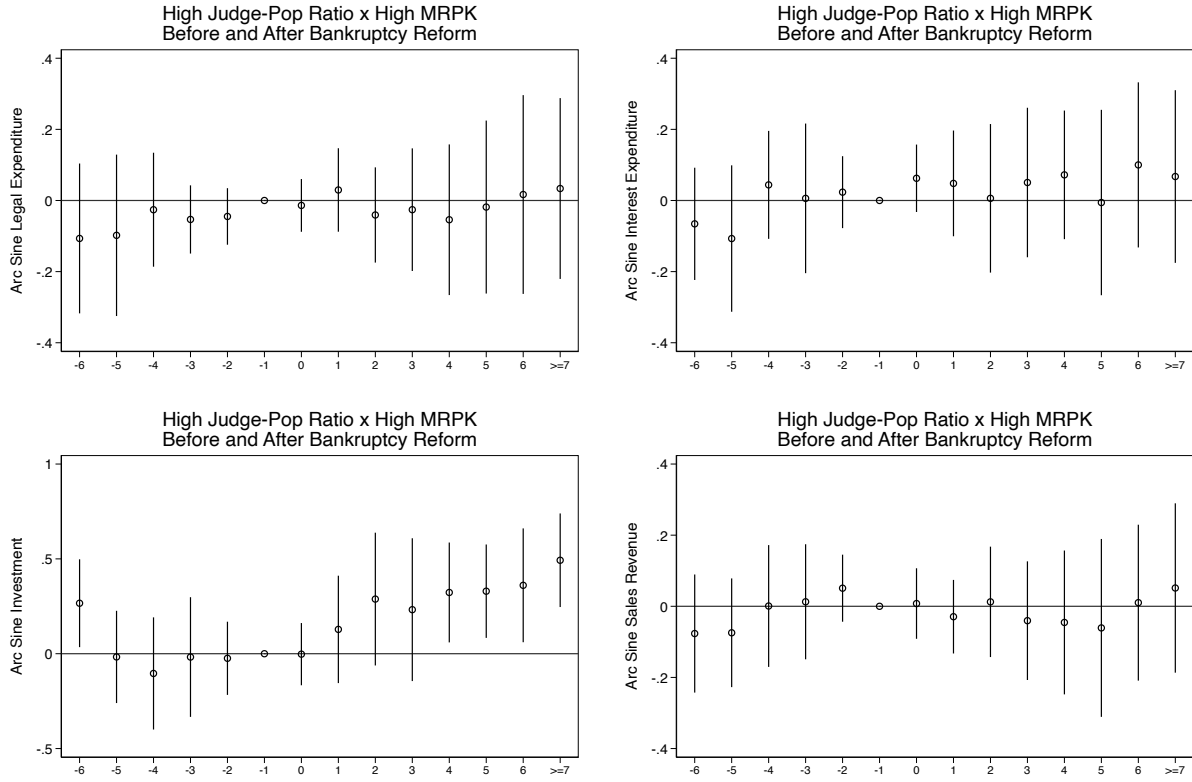
Notes: Panels A and B above present the estimates of event study implementation of [Equation 1](#) using borrowing (Panel A) and short-term liquidity (Panel B) outcomes as dependent variables for high-default-risk and high-MRPK interactions, respectively. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. High default risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) before 2010. High-MRPK firms are those with estimated pre-2010 MRPK above median of the firm-level sample. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table 2](#).

Figure 4: Economic Outcomes: By Baseline Default Risk



Notes: The figures above present the triple interaction estimates of event study implementation of [Equation 1](#) using financial and economic outcomes as dependent variables for firms classified as high-default risk in high judge-population ratio districts. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table 4](#).

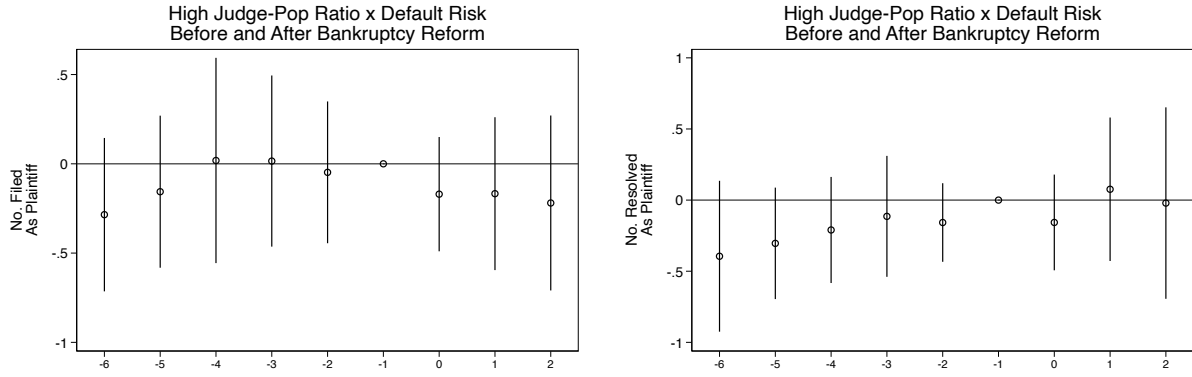
Figure 5: Economic Outcomes: By Baseline MRPK



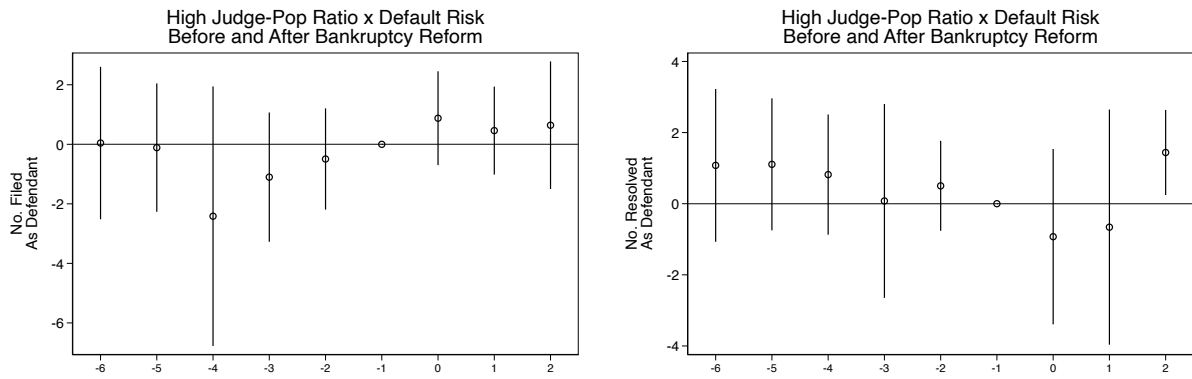
Notes: The figures above present the triple interaction estimates of event study implementation of [Equation 1](#) using financial and economic outcomes as dependent variables for firms classified as high-MRPK in high judge-population ratio districts. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table 5](#).

Figure 6: Litigation in Courts by Firms: By Baseline Default-Risk

Panel A: As Plaintiff



Panel B: As Defendant



Notes: The figures above present the triple interaction estimates of event study implementation of [Equation 1](#) using legal case-level outcomes in the corresponding district court as dependent variables for firms classified as high default risk in high judge-population ratio districts. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes. High judge to population ratio districts are those with the ratio above the sample average. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. Panel A presents the estimates on case-level outcomes when high-risk firms appear as plaintiff in a case. That is, when such firms initiate the suit in courts. Panel B presents the estimates when such firms appear as defendants in a case, i.e., when someone else files a suit against them. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table A.13](#).

10 Tables

Table 1: Summary Statistics

	(1) No. Firms	(2) Mean	(3) SD
Judge to Population Ratio (per million)	3582	51.74	49.60
Estimated MRPK	3381	9.09	41.76
Credit Rating (1 lowest-8 highest)	3582	5.78	1.78
Public Ltd.	3582	0.80	0.40
Age (years)	3580	40.82	19.36
Share Manufacturing	3582	0.59	0.49
Share Construction	3582	0.11	0.31
Share Trade/Retail	3582	0.19	0.39
Share Services	3582	0.02	0.15
Sales Rev (Million INR)	3411	4605.24	25855.26
Total Assets (Million INR)	3582	5961.70	31772.94

Notes: The main sample includes over 3000 formal, non-financial sector firms registered in districts with courts classified as high or low judicial capacities (judge to population ratio) that form the main population of interest for this study. Note that the judge to population ratio in this sample is different from the national average of 18.6, which includes districts with no registered formal sector firms. Age is calculated as the difference between current year and the year of firm's incorporation. The remaining 8% of sample firms belong to other non-financial sectors represented by industrial classification (NIC) codes that could not be neatly binned into the sector categories above. The average sales revenue and asset value are computed for period prior to the reform.

Table 2: Bankruptcy Reform and District Judicial Capacity: Firms' Borrowing and Liquidity

	(1) Arc Sine Borrowing	(2) Arc Sine Net Liquidity	(3) Arc Sine Borrowing	(4) Arc Sine Net Liquidity
High Jdg Pop Ratio x High Default Risk x Post	-0.156** (0.0678)	0.818*** (0.299)		
High Default Risk x Post	0.0856 (0.0675)	-0.608*** (0.216)		
High Jdg Pop Ratio x High MRPK x Post			-0.00876 (0.0532) [0.189]	0.331 (0.261) [0.424]
High MRPK x Post			0.0609 (0.0487)	-0.531** (0.225)
Observations	57840	57885	47970	48000
No. Districts	164	164	150	150
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	7.400	2.360	7.480	2.060
SD Dep Var	2.200	6.520	2.290	6.700
Adj R-Squared	0.801	0.624	0.797	0.616

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: Col 1-2 above present the estimates from Equation 1 using borrowing and short-term liquidity outcomes as dependent variables for high-default-risk firm interactions. Col 3-4 present the estimates for high-MRPK firm interactions. “Post” is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. High-judge-to-population ratio districts are those with the ratio above the sample average. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Standard errors are clustered by district and reported in parentheses. Boot-strapped standard errors for the MRPK triple difference estimate reported in [.]. The event study figures are in Figure 3.

Table 3: Firm Default Risk and MRPK

	(1)	(2)	(3)
	High MRPK	High MRPK	High MRPK
High Default Risk	-0.0918*** (0.0266)	-0.0404 (0.0265)	-0.0460 (0.0281)
Constant	0.482*** (0.00883)	0.474*** (0.00772)	0.478*** (0.00788)
Observations	3582	3431	3347
Fixed Effect	None	Industry	Industry, District
Adj R-Squared	0.00291	0.257	0.259

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table reports firm-level correlations between default risk and their high-MRPK classification using data from pre-reform period. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Col 1 presents simple, bi-variate correlation. Cols 2-3 include location and industry fixed effects to account for risk or productivity factors driven either by location of registration or industry. Heteroskedasticity robust standard errors reported in parentheses.

Table 4: Bankruptcy Reform and District Judicial Capacity: Firms' Production Outcomes by Default Risk

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
High Jdg Pop Ratio x High Default Risk x Post	-0.112*** (0.0398)	-0.234*** (0.0538)	-0.365* (0.200)	-0.0982 (0.0742)
High Default Risk x Post	-0.0224 (0.0407)	0.0401 (0.0458)	0.0283 (0.0845)	-0.0337 (0.0593)
Observations	50190	57210	44700	57360
No. Districts	152	164	145	163
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.870	5.050	5.330	8.650
SD Dep Var	1.720	1.830	2.920	1.870
Adj R-Squared	0.906	0.805	0.878	0.871

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from [Equation 1](#) using legal and interest expenditures, long-run investment (maturity > 1 year), and sales revenue as dependent variables for high-default-risk firm interactions. "Post" is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. High-judge-to-population ratio districts are those with the ratio above the sample average. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) in the period prior to the bankruptcy reform. Standard errors are clustered by district. The event study figures are in [Figure 4](#).

Table 5: Bankruptcy Reform and District Judicial Capacity: Firms' Production Outcomes by MRPK

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
High Jdg Pop Ratio x High MRPK x Post	0.0458 (0.0618) [0.138]	0.0560 (0.0607) [0.219]	0.316*** (0.0707) [0.164]	0.00498 (0.0495) [0.094]
High MRPK x Post	-0.0398 (0.0278)	-0.0310 (0.0440)	-0.0464 (0.0713)	-0.132*** (0.0401)
Observations	41070	47400	36795	47520
No. Districts	136	150	130	149
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.910	5.140	5.540	8.690
SD Dep Var	1.760	1.890	2.950	1.940
Adj R-Squared	0.905	0.801	0.877	0.873

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from [Equation 1](#) using legal and interest expenditures, long-run investment (maturity > 1 year), and sales revenue as dependent variables for high-MRPK firm interactions. "Post" is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. High-judge-to-population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Standard errors are clustered by district and reported in parentheses. Boot-strapped standard errors for the MRPK triple difference estimate reported in [.]. The event study figures are in [Figure 5](#).

Table 6: Bankruptcy Filings under IBC

	(1)	(2)	(3)	(4)	(5)	(6)
	In Bankruptcy	Restructuring	Liquidation	In Bankruptcy	Restructuring	Liquidation
High Risk x Judge-Pop Ratio	-0.000393* (0.000226)	-0.0000469 (0.000143)	-0.000346** (0.000156)			
High MRPK x Judge-Pop Ratio				-0.0000422 (0.000123)	-0.0000362 (0.0000930)	-0.00000591 (0.0000707)
Judge-Pop Ratio	-0.000469 (0.000370)	-0.000504 (0.000354)	0.0000347 (0.000151)	-0.000439 (0.000371)	-0.000363 (0.000302)	-0.0000761 (0.000204)
High Risk Firm	0.0638*** (0.0205)	0.0221 (0.0159)	0.0418** (0.0169)			
High MRPK Firm				0.00977 (0.0165)	0.000646 (0.00995)	0.00912 (0.0103)
Observations	4947	4947	4947	4234	4234	4234
No. Districts	249	249	249	242	242	242
State FE	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y
Mean Dep Var	0.0700	0.0300	0.0300	0.0700	0.0400	0.0300
SD Dep Var	0.250	0.180	0.180	0.260	0.200	0.180
Adj R-Squared	0.0335	0.00967	0.0363	0.0400	0.0130	0.0444

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above examines bankruptcy filings under the IBC, 2016 using claims data from the Insolvency and Bankruptcy Board of India. The firms filed for bankruptcy are then matched to the Prowess sample by their CIN code, a primary key available in both datasets for the merge. The dependent variables are whether the Prowess firm is found in the bankruptcy filing data (Col 1, 4), an indicator for restructuring outcome coded as 0 for those not under bankruptcy (Col 2, 5), and an indicator for liquidation outcome coded as 0 for those not under bankruptcy (Col 3, 6). Since the bankruptcy data is only available after the implementation of IBC, the table examines differential effects of better district judicial capacity (measured as judge to population ratio) across firms by their default risk and MRPK. High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's). High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Standard errors are clustered by district.

Online Appendix

A.1 A Conceptual Framework on Credit Allocation

I sketch a simple two-period model on credit allocation in an environment where bank officials' incentives are based on their recognition of bad loans (NPA). The status quo creates an incentive against detection and reporting of loan defaults by the bank officials, who continue to provide fresh loans to finance the past, defaulted loan ("ever-greening"). The bankruptcy reform attempts to change this incentive. This framework extends the model in [Banerjee and Duffo \(2014\)](#) by making ever-greening by a bank official a function of both the bankruptcy regime and local judicial capacity.

Consider a representative bank official, who has L units of credit to allocate to borrowers in each period. Borrowers are of two types in the population - a type H with a share ϕ in the population and a type L constituting the remaining $1 - \phi$. The type H borrower has a deterministic production function with probability of success, $s_H = 1$ whereas the type L face a non-zero probability of production failure $1 - s$ such that $s_L = s < 1$. The borrower type is unknown to both bank official and borrower in the initial period, which gets revealed with the outcome of production. So in the first round of allocation, the official equally distributes total credit L equally among the borrowers such that each borrower gets l units of credit.

At the end of period 1, the official receives a signal that the borrower is of type H conditional on observed production success. This signal probability is $S_H = \frac{\phi}{\phi + (1-\phi)s}$. Among type L borrowers, $(1 - \phi)(1 - s)$ fail. At the beginning of period 2, the official can either declare loan provided to type L as default and face a penalty $P_1(l, \Gamma)$ or provide fresh loan to type L borrower to continue production, which would lead to a bigger default at the end of period 2 with probability $1 - s$. Penalty in period 1 is a function of loan size, l , and local court quality Γ , such that $\frac{\partial P_1}{\partial \Gamma} > 0$, i.e. well-functioning courts generate a larger penalty, and $\frac{\partial P_1}{\partial l} > 0$. The official will bail out type L borrower by lending l_L in period 2 if $sf(l_L - l) \geq l_L$, where l due from period 1 is paid out of l_L and the rest put into production. Rationality implies that $l_L^* = sf(l_L^* - l)$, that is, the official lends the minimum amount to type L borrower to hedge the risk of default at the end of period 2. Therefore, the official compares the option of bailing out at the beginning of period 2 to declaring a default, which costs him $P_1(l, \Gamma)$. So, bailing will be the dominant strategy if the penalty from default at the end of period 2 is less than the penalty from default at the end of period 1. Penalty in period 2 has an additional institutional parameter, Θ - a measure of creditor (bank's) rights in the form of redressal through bankruptcy process. I assume that $\frac{\partial P_2}{\partial \Gamma} > 0$, $\frac{\partial P_2}{\partial \Theta} > 0$ and $\frac{\partial^2 P_2}{\partial \Gamma \partial \Theta} > 0$. That is, penalty

in period 2 is higher due to the complementary effects of default detection through courts as well as protection of creditor rights under bankruptcy reform. Thus, bailing will be the optimal response if the following holds:

$$(1-s)P_2(l_L^*, \Theta, \Gamma) \leq P_1(l, \Gamma)$$

$$P_2(l_L^*, \Theta, \Gamma) \leq \frac{P_1(l, \Gamma)}{1-s}$$

Assuming that the penalty can be expressed as a share of the loan amount, $P_1(l, \Gamma) = lP_1(\Gamma)$ and $P_2(l_L^*, \Theta, \Gamma) = l_L^*P_2(\Theta, \Gamma)$. Therefore, the bailout loan can be expressed as:

$$l_L^* \leq \frac{lP_1(\Gamma)}{(1-s)P_2(\Theta, \Gamma)}$$

The right hand side of the above expression for period 2 bailout loan varies by both the quality of local courts, Γ , as well as the overall environment of creditor rights Θ . The bankruptcy reform increases Θ whereas higher judicial capacity (higher judge to population ratio) increases Γ . From this set-up, I get the following results with corresponding hypotheses to test with data.

Result 1: The marginal reduction in bailout loan, l_L^* , is amplified by increasing institutional parameters Θ and Γ , i.e. $\frac{\partial l_L^*}{\partial \Theta \partial \Gamma} < 0$. That is, bailout lending to type L borrowers decreases in credit markets (districts) with higher judicial capacity after the bankruptcy reform.

Proof¹

$$\frac{\partial l_L^*}{\partial \Theta \partial \Gamma} = \frac{l}{1-s} \left(\frac{-1}{P_2^2} \frac{\partial P_1}{\partial \Gamma} \frac{\partial P_2}{\partial \Theta} - \frac{P_1}{P_2^2} \frac{\partial^2 P_2}{\partial \Theta \partial \Gamma} + \underbrace{\frac{2P_1}{P_2^3} \frac{\partial P_2}{\partial \Gamma} \frac{\partial P_2}{\partial \Theta}}_{\rightarrow 0} \right) < 0$$

Result 2: In the absence of strong creditor rights through the bankruptcy reform, the effect of higher judicial capacity on bailout loan is ambiguous or likely positive.

¹The cubic for a large positive penalty P_2 is orders of magnitude large relative to P_1 , which makes the ratio a very small number such that the third term in the expression tends to 0 in limit.

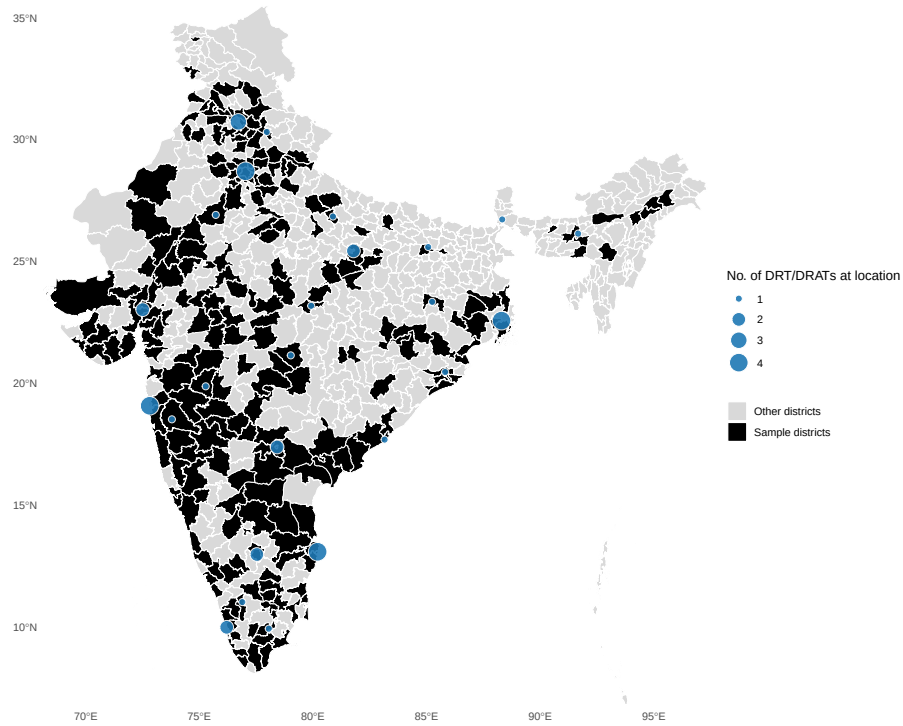
Proof

$$\frac{\partial l_L^*}{\partial \Gamma} = \frac{l}{1-s} \left(\frac{1}{P_2} \frac{\partial P_1}{\partial \Gamma} - \frac{P_1}{P_2^2} \frac{\partial P_2}{\partial \Gamma} \right) > \text{ or } < 0$$

Implication Given this framework, the following section presents the empirical strategy to test the hypotheses on credit allocation generated by the above results. Specifically, from result 1, I examine whether lending to “riskier” borrowers (from the perspective of default probability) diminishes after the bankruptcy reform in districts with better judicial capacity. Result 2 suggests plausible “parallel trends” prior to the reform since lending decisions to specific type of borrower are more likely to be driven by interaction with bank official’s incentives rather than institutional quality alone.

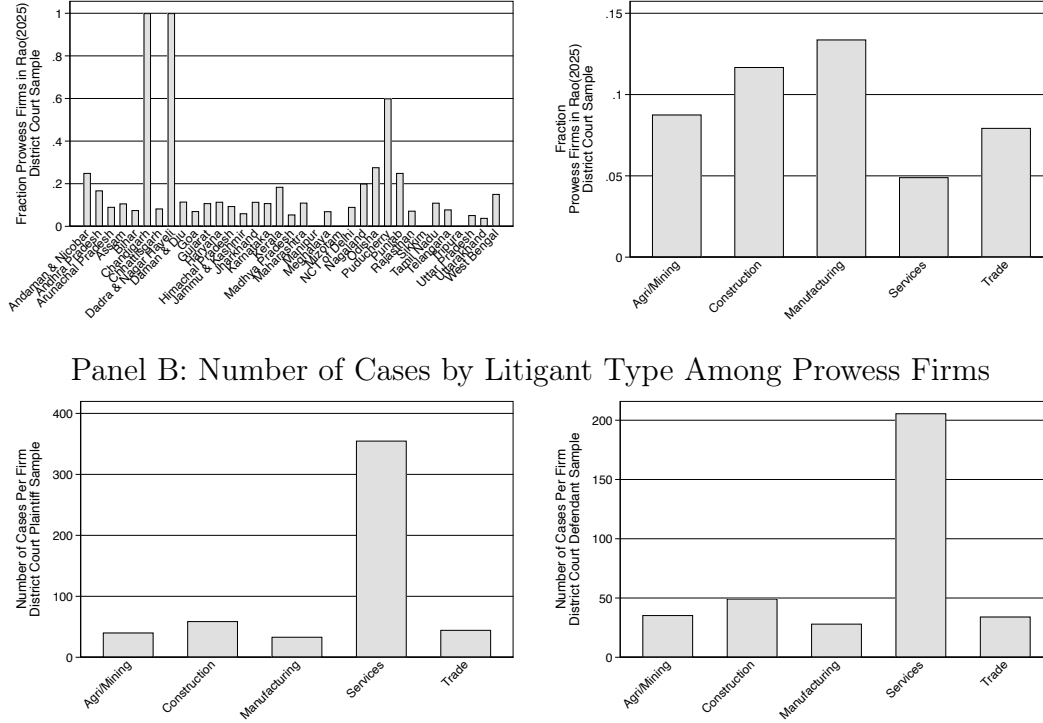
Additional figures and tables

Figure A.1: Sample Districts and DRT/NCLT Location



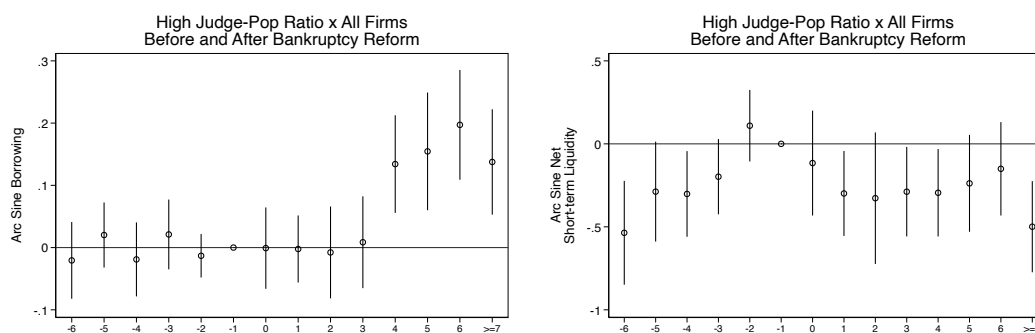
Notes: The above map shows the spatial distribution of district courts in the matched-firm sample and DRT locations. There are 243 unique district courts in the sample whereas DRTs are concentrated only in 26 unique locations, which overlap with the district court sample as well as the location of NCLT benches.

Figure A.2: Match Between Prowess Firms and Litigating Firms in District Courts
Panel A: Prowess Firm Coverage by State and Sector



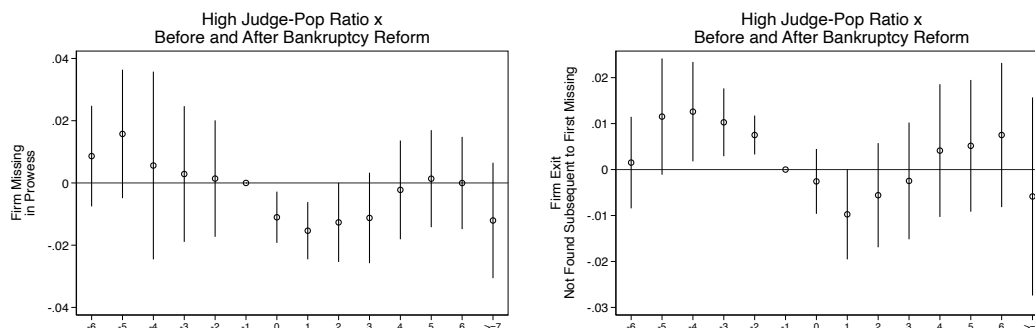
Notes: The above map shows the spatial and sectoral distribution of non-financial Prowess firms found in the district court case-level data from [Rao \(2025\)](#). Panel A shows the distribution of Prowess firms in the case-level data by the state of their registered office location and sector, respectively. This covers about 11% of Prowess firms on average. Panel B shows distribution of cases by Prowess firm sector (number of cases per firm) separately by their status as a plaintiff or defendant in the case-level data.

Figure A.3: Borrowing/Liquidity effects of bankruptcy reform on all firms by district judicial capacity

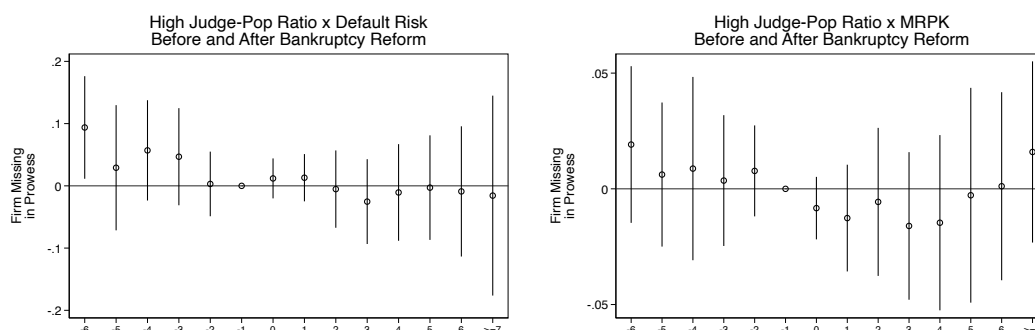


Notes: The figures report the event study estimates of bankruptcy reform on all firms by comparing those registered in high-judicial-capacity districts, measured using baseline judge-to-population ratio, with those registered in low capacity districts using borrowing and short-run liquidity as outcome variables. The error bars denote 95% confidence intervals and standard errors are clustered by district. These specifications are used to test the trends in borrowing and liquidity outcomes of firms in the period before the reforms.

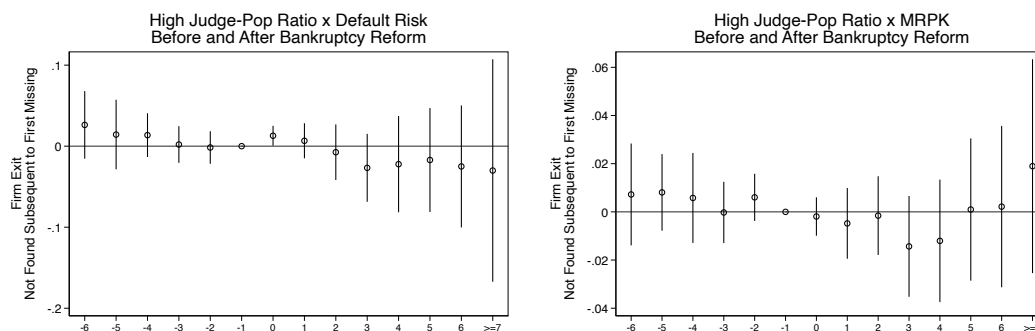
Figure A.4: Unobserved Firm Exits: Missing Financial Variables in Prowess Data Over Time
Panel A: DiD using All Firms



Panel B: Missing Reporting (Triple-D) By Default-Risk or MRPK

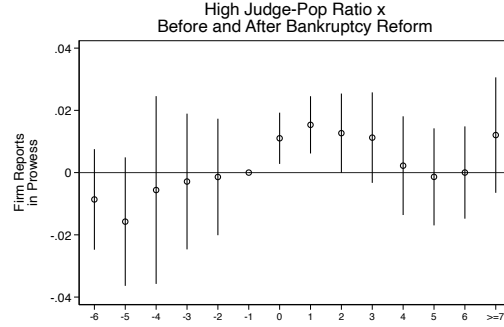


Panel C: Permanent Missing (Triple-D) By Default-Risk or MRPK

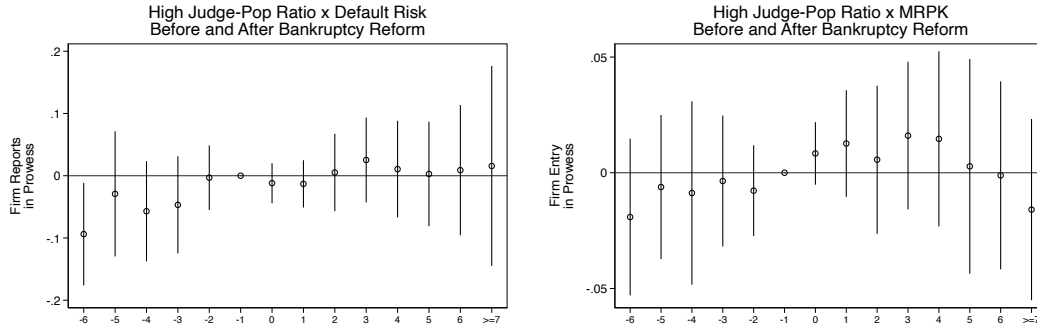


Notes: The figures depict the effect of the reform on firm disappearance in data using all Prowess firms in the sample districts. Formal firm exits through liquidations are unobserved in the data. I code firm disappearance in data along two margins using Prowess. First, I measure whether a firm does not report their financial data for any year in the sample time period 2010-2024. I call this “Firm Missing” to indicate years where I don’t observe the corresponding financial data for the firm. Second, I measure whether a firm completely stops reporting data after the first time they stop reporting. This indicates whether the firm exits the Prowess dataset. Panel A presents event study using these measures as the dependent variable in DiD design comparing the outcome between high judicial capacity districts with those in low capacity districts. Panel B and Panel C present the triple difference event study results by firms classified as high default risk or high MRPK, respectively, and the underlying judicial capacities of the district. All specifications include firm fixed effects, and industry-year fixed effects and the triple difference specifications additionally use district-year fixed effects. The error bars denote 95% confidence intervals and standard errors are clustered by district.

Figure A.5: Selection into Reporting: Financial Variables in Prowess Data Over Time
Panel A: DiD using All Firms

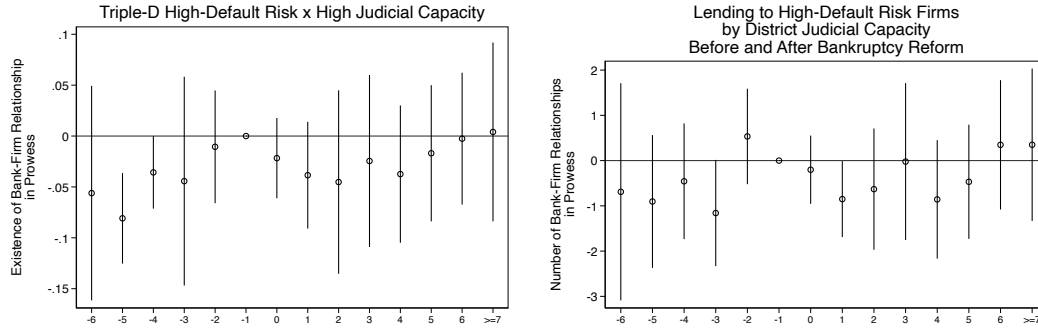


Panel B: Triple-D By Default-Risk or MRPK

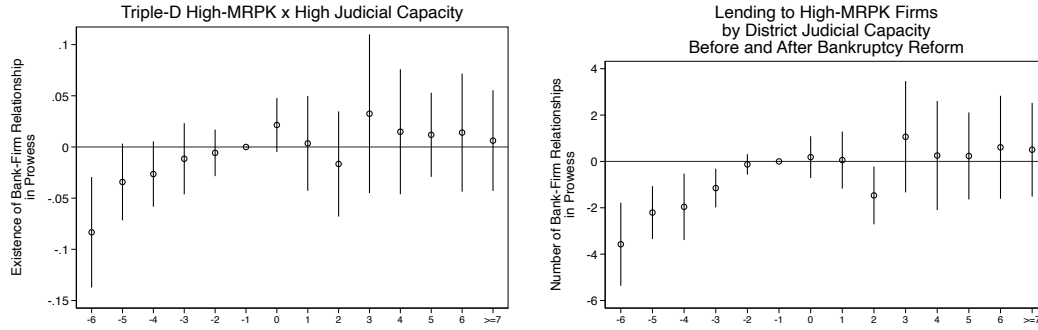


Notes: The figures depict the effect of the reform on firms' reporting of financial variables (non-missingness) in data using all Prowess firms in the sample districts. I measure whether a firm reports their financial data for each year in the sample time period 2010-2024. I call this "Firm Reports" to indicate years where I do observe the corresponding (non-missing) financial data for the firm. Panel A presents event study using this measure as the dependent variable in DiD design comparing the outcome between high judicial capacity districts with those in low capacity districts. Panel B presents the triple difference event study results by firms classified as high default risk or high MRPK, respectively, and the underlying judicial capacities of the district. All specifications include firm fixed effects, and industry-year fixed effects and the triple difference specifications additionally use district-year fixed effects. The error bars denote 95% confidence intervals and standard errors are clustered by district.

Figure A.6: Firm-Bank Relationship
Panel A: Triple-D By Default-Risk

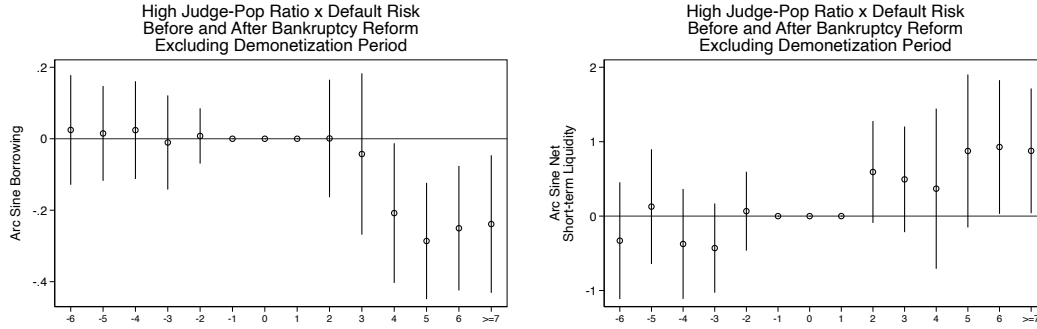


Panel B: Triple-D By MRPK

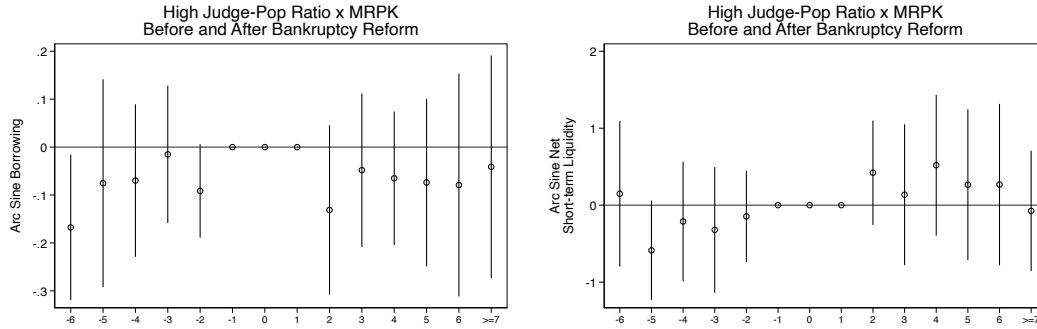


Notes: The figures depict the effect of the reform on firm-bank relationship observed in the “banker” table corresponding to the Prowess firm sample. There are two relevant margins of this relationship: first, whether there exists a relationship between a firm and a bank and second, total number of banking relationships by a firm. There are 658 unique banks with at least one relationship with the sample firm. The banks span commercial banks as well as regional banks and credit unions. The dependent variable “Existence of Bank-Firm Relationship” is a dummy variable that takes value 1 when I observe a relationship between a specific bank and a sample firm for a given year and 0 otherwise. The variable “Number of Bank-Firm Relationships” is constructed by aggregating the existence indicator across all banks for a specific firm for a given calendar year. The data includes a balanced panel of all sample firms over the study period 2010-2024. Additionally, the existence indicator also includes the specific banking partner for the firm. For regressions using the existence variable, I also include bank fixed effects in addition to firm and district-year fixed effects. Regressions using the aggregate variable includes only firm and district-year fixed effects. Panels A and B present the triple difference event study results by firms classified as high default risk or high MRPK, respectively, and the underlying judicial capacities of the district. The error bars denote 95% confidence intervals and standard errors are clustered by district.

Figure A.7: Robustness to other concurrent shocks: Demonetization, GST
Panel A: High Default-Risk Firms

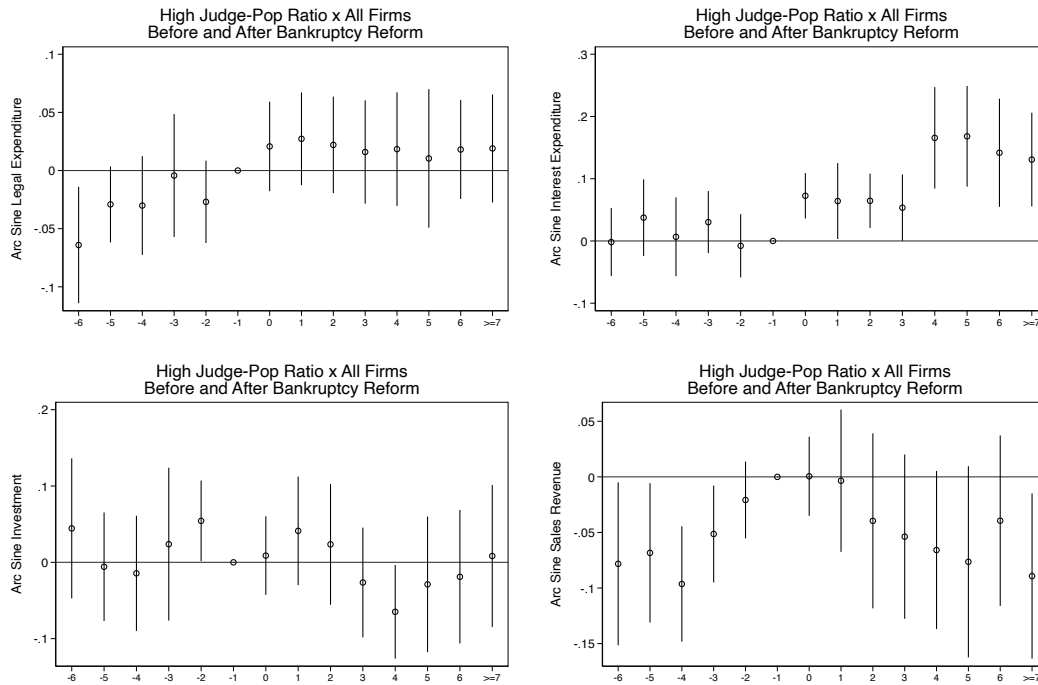


Panel B: High MRPK Firms



Notes: Panels A and B above present the estimates of event study implementation of [Equation 1](#) using borrowing (Panel A) and short-term liquidity (Panel B) outcomes as dependent variables for high-default-risk and high-MRPK interactions, respectively. These figures exclude the years 2016 and 2017 with many, concurrent national policy shocks such as demonetization in 2016 and the roll-out of GST in 2017. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. High default risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody's) before 2010. High-MRPK firms are those with estimated pre-2010 MRPK above median of the firm-level sample. Error bars present 95% confidence interval. Standard errors are clustered by district.

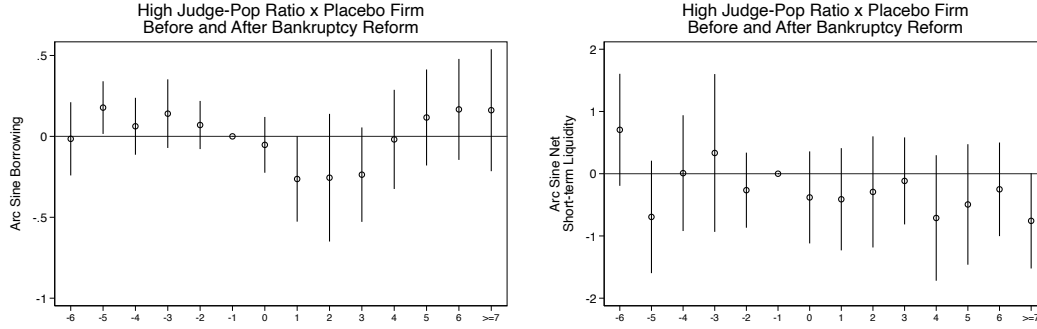
Figure A.8: Balance sheet effects of bankruptcy reform on all firms by district judicial capacity



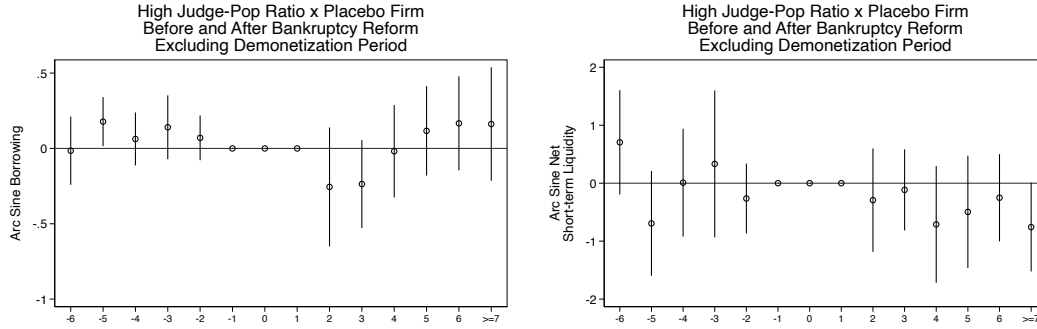
Notes: The figures report the event study estimates of bankruptcy reform on all firms by comparing those registered in high-judicial-capacity districts, measured using baseline judge-to-population ratio, with those registered in low capacity districts using balance sheet outcome variables including legal and interest expenditures, long run investments and sales revenue. The error bars denote 95% confidence intervals and standard errors are clustered by district. These specifications are used to test the trends in outcomes of firms in the period before the reforms.

Figure A.9: Placebo Firm Analysis

Panel A: Borrowing and Liquidity Effects Among Placebo Firms

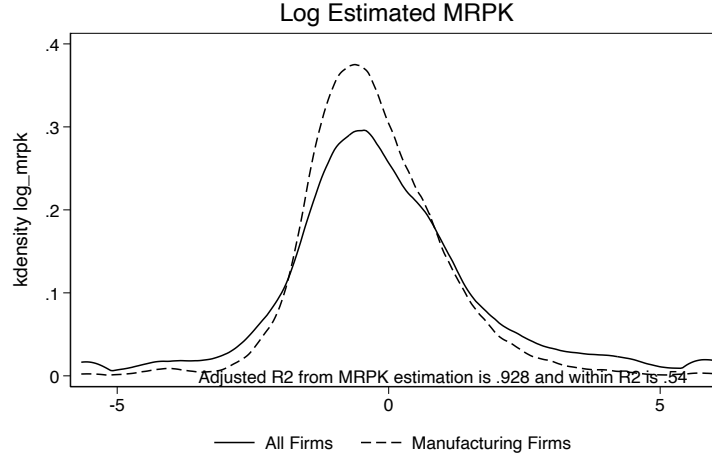


Panel B: Excluding Demonetization/GST Years

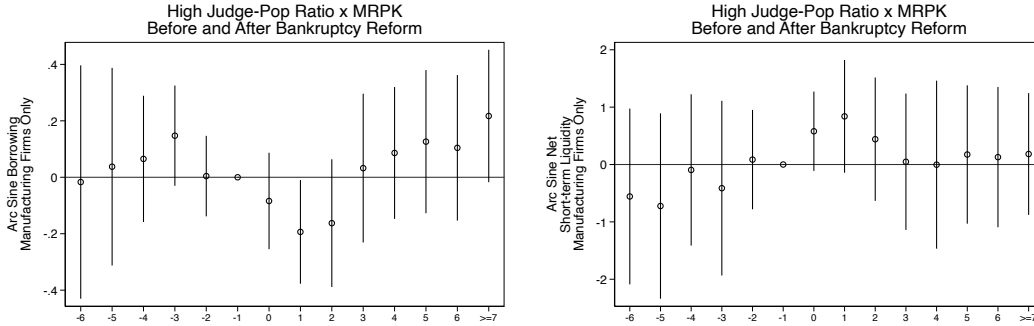


Notes: Panel A presents the estimates of event study implementation of [Equation 1](#) using borrowing and short-term liquidity outcomes as dependent variables for placebo-firm district judicial capacity interactions, respectively. Placebo firms are those with perfect credit ratings in the 2005-2010 time period. Credit allocation to such firms should not theoretically be affected by institutional quality or lenders' ever-greening motives incentives. Figures in Panel B additionally exclude the years 2016 and 2017 with many, concurrent national policy shocks such as demonetization in 2016 and the roll-out of GST in 2017. In all figures, the reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. High judge to population ratio districts are those with the ratio above the sample average. Error bars present 95% confidence interval. Standard errors are clustered by district.

Figure A.10: MRPK Estimation and Accounting for Measurement Error
Panel A: Estimated MRPK Distribution



Panel B: Borrowing and Liquidity Effects by MRPK in Manufacturing



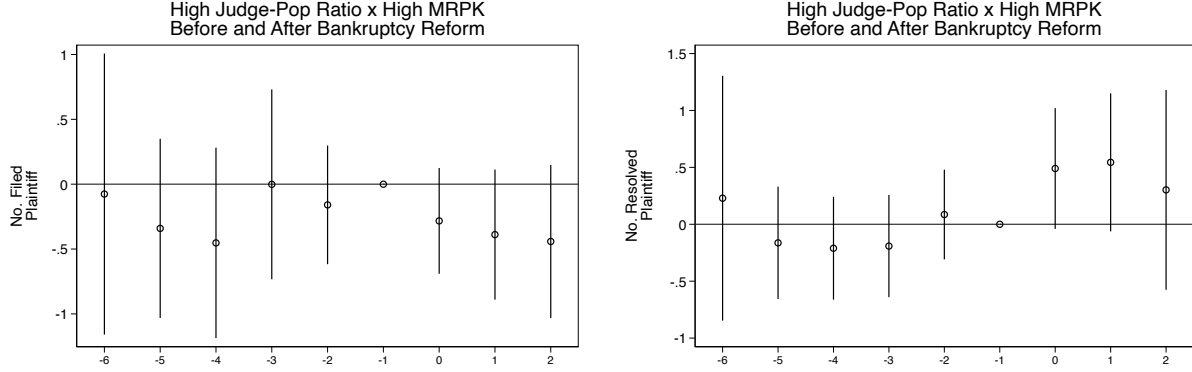
Notes: Panel A presents the kernel density distribution of Log estimated MRPK using the following procedure:

$$y_{it} = \beta_k K_{it} + \beta_l L_{it} + \phi_i + \phi_t + \varepsilon_{it}$$

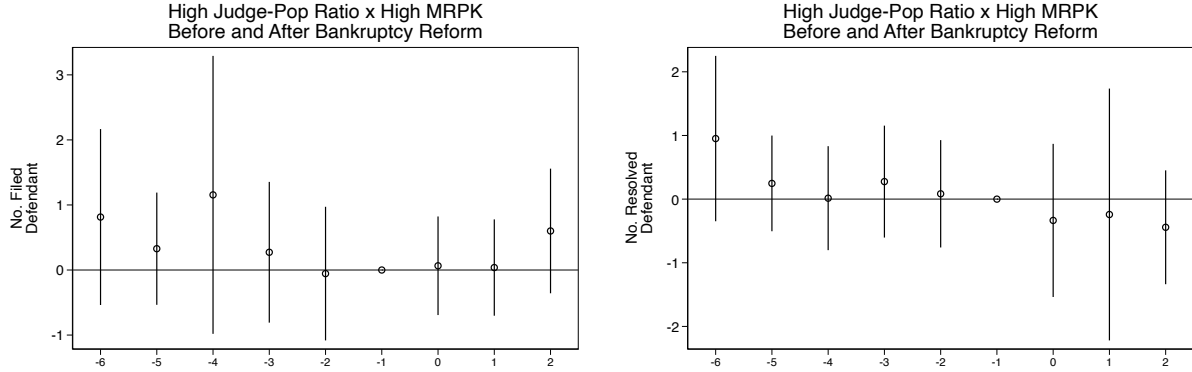
where y_{it} is log sales revenue, K_{it} is the log value of plant and machinery, L_{it} is log wagebill. Data from before 2010 is used for estimating MRPK as: $\widehat{MRPK} = \beta_k \times \frac{\text{sales revenue}}{\text{value of plant/mach}}$. Panel B presents the borrowing and liquidity effects in the high-MRPK triple difference estimation when the industry is limited to manufacturing only, where capital is more reliably measured. Error bars present 95% confidence interval and standard errors are clustered by district.

Figure A.11: Litigation in Courts by Firms: By Baseline MRPK

Panel A: As Plaintiff

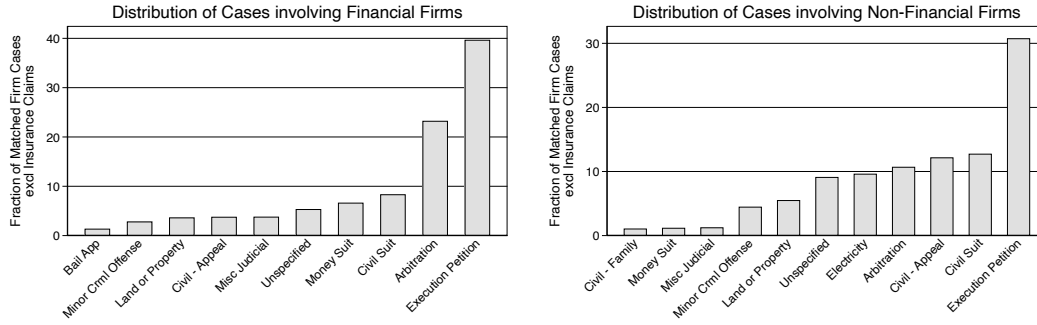


Panel B: As Defendant

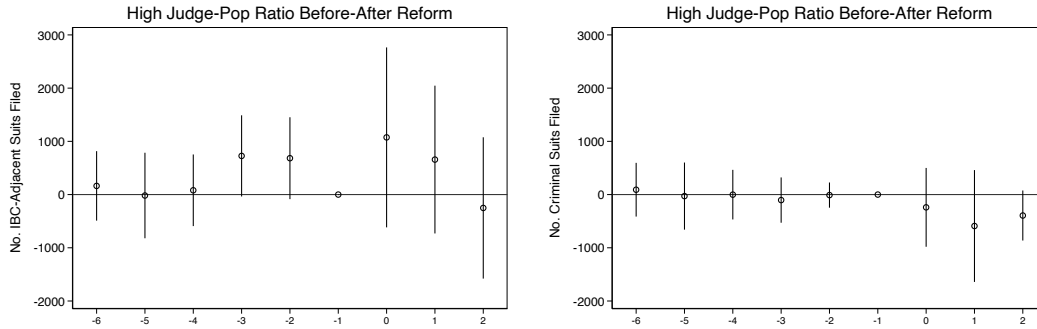


Notes: The figures above present the triple interaction estimates of event study implementation of [Equation 1](#) using legal case-level outcomes in the corresponding district court as dependent variables for firms classified as high-MRPK in high judge-population ratio districts. The reference year is 2015, a year before the bankruptcy bill becomes an act of the Indian Parliament. The coefficients are all relative to the reference year. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes. High judge to population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Panel A presents the estimates on case-level outcomes when high-MRPK firms appear as plaintiff in a case. That is, when such firms initiate the suit in courts. Panel B presents the estimates when such firms appear as defendants in a case, i.e., when someone else files a suit against them. Error bars present 95% confidence interval. Standard errors are clustered by district. The DDD estimates are reported in [Table A.14](#).

Figure A.12: Case Type Distribution
Panel A: Case Type Distribution in Matched Firm Data



Panel B: IBC-Adjacent vs Non-Adjacent Cases in High Judge Capacity Over Time



Notes: Panel A presents percentage of cases by type in the matched-firm case-level data from [Rao 2025](#). [Ash et al. 2025](#) does not provide the identity of litigants to generate matched-firm case-level data. Panel B uses case-level data from [Ash et al. 2025](#) to calculate the number of IBC-adjacent such as money suits, recovery suits, debt arbitration, challenges to possession, execution petitions, and placebo (criminal) cases over time. The graph presents event-study estimates of IBC reform by district judicial capacity in a DiD specification with district, year, and state-year fixed effects. Standard errors are clustered by district.

Table A.1: Firm Type Across High and Low Judge-Pop Ratio Districts

Panel A							
	High Default Risk				Low Default Risk		
	High Judge-Pop	Low Judge-Pop	P-val	High Judge-Pop	Low Judge-Pop	P-val	
Overall Credit Rating	2.321 (0.916)	2.342 (0.897)	0.763	6.313 (1.399)	6.009 (1.354)	0.006	
Debt to Asset	0.459 (0.351)	0.570 (0.441)	0.929	0.375 (0.342)	0.401 (0.330)	0.319	
Total Assets (Deflated)	3341.437 (5561.150)	2468.327 (3934.879)	0.148	30554.326 (164600.758)	13844.836 (45273.843)	0.000	
<i>N</i>	159	374		1643	2971		
Panel B							
	High MRPK				Low MRPK		
	High Judge-Pop	Low Judge-Pop	P-val	High Judge-Pop	Low Judge-Pop	P-val	
Overall Credit Rating	6.091 (1.719)	5.862 (1.641)	0.758	5.972 (1.857)	5.575 (1.812)	0.095	
Debt to Asset	0.319 (0.269)	0.349 (0.281)	0.006	0.514 (0.484)	0.526 (0.458)	0.850	
Total Assets (Deflated)	20241.117 (103905.870)	10511.783 (33645.469)	0.000	46931.853 (227576.917)	17167.062 (54591.554)	0.001	
<i>N</i>	740	1162		833	1702		

Notes: This table presents debt-related characteristics of high and low default risk firms (Panel A) and high and low MRPK firms (Panel B) in high and low judge-population ratio districts using data from pre-2010. Overall credit rating ranges from 1 to 8, with 1 indicating lowest possible rating and 8 indicating highest. Debt to Asset is total debt (borrowing) to total asset ratio and total asset is measured in Million INR in 2015 levels.

Table A.2: Unobserved Firm Exits: Missing Financial Variables in Prowess Data Over Time

	(1) Firm Missing Prowess All	(2) Firm Exit Prowess All	(3) Firm Missing Prowess Default-Risk	(4) Firm Exit Prowess Default-Risk	(5) Firm Missing Prowess MRPK	(6) Firm Exit Prowess MRPK
High Jdg Pop Ratio x Post	-0.0113 (0.00903)	-0.00743 (0.00633)	0 (.)	0 (.)	0 (.)	0 (.)
High Jdg Pop Ratio x High Default Risk x Post			-0.0434* (0.0236)	-0.0286 (0.0281)		
High Default Risk x Post			0.0187 (0.0196)	0.0429*** (0.0146)		
High Jdg Pop Ratio x High MRPK x Post					-0.00764 (0.0159)	-0.00251 (0.0132)
High MRPK x Post					-0.0123 (0.0125)	0.00378 (0.0130)
Observations	64275	64275	63060	63060	63060	63060
No. Districts	242	242	165	165	165	165
Firm FE	X	X	X	X	X	X
District-Year FE	No	No	X	X	X	X
Industry-Year FE	X	X	X	X	X	X
Mean Dep Var	0.310	0.0300	0.310	0.0300	0.310	0.0300
SD Dep Var	0.460	0.170	0.460	0.170	0.460	0.170
Adj R-Squared	0.772	0.699	0.774	0.701	0.774	0.701

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table is the DiD/DDD equivalent to [Figure A.4](#) using firm disappearance in the financial data as the dependent variable. I use all firms observed in Prowess data on or before 2010 and generate indicators for missing financial data to assess whether the interaction between bankruptcy reform and local district judicial capacity affects reporting compliance by firms. Column 1 reports DiD estimates examining the interaction between high judicial capacity and the timing of bankruptcy reform using all firms. Columns 2 and 3 report DDD estimates using the interaction between firm characteristic, district judicial capacity, and the timing of the reform. Standard errors are clustered by district.

Table A.3: Selection into Reporting: Financial Variables in Prowess Data Over Time

	(1) Firm Reports Prowess All	(2) Firm Reports Prowess Default-Risk	(3) Firm Reports Prowess MRPK
High Jdg Pop Ratio x Post	0.0113 (0.00903)	0 (.)	0 (.)
High Jdg Pop Ratio x High Default Risk x Post		0.0434* (0.0236)	
High Default Risk x Post		-0.0187 (0.0196)	
High Jdg Pop Ratio x High MRPK x Post			0.00764 (0.0159)
High MRPK x Post			0.0123 (0.0125)
Observations	64275	63060	63060
No. Districts	242	165	165
Firm FE	X	X	X
District-Year FE	No	X	X
Industry-Year FE	X	X	X
Mean Dep Var	0.690	0.690	0.690
SD Dep Var	0.460	0.460	0.460
Adj R-Squared	0.772	0.774	0.774

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table is the DiD/DDD equivalent to [Figure A.4](#) using firm disappearance in the financial data as the dependent variable. I use all firms observed in Prowess data on or before 2010 and generate indicators for missing financial data to assess whether the interaction between bankruptcy reform and local district judicial capacity affects reporting compliance by firms. Column 1 reports DiD estimates examining the interaction between high judicial capacity and the timing of bankruptcy reform using all firms. Columns 2 and 3 report DDD estimates using the interaction between firm characteristic, district judicial capacity, and the timing of the reform. Standard errors are clustered by district.

Table A.4: Bank-Firm Relationship

	DiD (1) Bank-Firm Relation	(2) No. Banks	Triple-D (3) Bank-Firm Relation	(4) No. Banks	Triple-D (5) Bank-Firm Relation	(6) No. Banks
High Jdg Pop Ratio x Post	0.00249 (0.0125)	-0.327 (0.265)	0 (.)	0 (.)	0 (.)	0 (.)
High Jdg Pop Ratio x High Default-Risk x Post			0.0160 (0.0297)	0.188 (0.622)		
High Default-Risk x Post			-0.0171 (0.0249)	0.814* (0.433)		
High Jdg Pop Ratio x High MRPK x Post					0.0291 (0.0264)	1.482 (1.120)
High MRPK x Post					0.00530 (0.0185)	-0.0915 (0.602)
Observations	343620	343620	343290	343290	343290	343290
No. Districts	236	236	214	214	214	214
Firm FE	X	X	X	X	X	X
District-Year FE	N	N	X	X	X	X
Bank FE	X	X	X	N	X	N
Mean Dep Var	0.380	6.760	0.380	6.760	0.380	6.760
SD Dep Var	0.490	8.370	0.490	8.370	0.490	8.370
Adj R-Squared	0.240	0.666	0.253	0.695	0.253	0.695

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table is equivalent to [Figure A.6](#), depicting the effect of the reform on firm-bank relationship observed in the “banker” table corresponding to the Prowess firm sample. There are two relevant margins of this relationship: first, whether there exists a relationship between a firm and a bank and second, total number of banking relationships by a firm. There are 658 unique banks with at least one relationship with the sample firm. The banks span commercial banks as well as regional banks and credit unions. The dependent variable “Existence of Bank-Firm Relationship” is a dummy variable that takes value 1 when I observe a relationship between a specific bank and a sample firm for a given year and 0 otherwise. The variable “Number of Bank-Firm Relationships” is constructed by aggregating the existence indicator across all banks for a specific firm for a given calendar year. The data includes a balanced panel of all sample firms over the study period 2010-2024. Additionally, the existence indicator also includes the specific banking partner for the firm. Col 1-2 present DiD estimates comparing the outcomes between high judicial capacity districts with those in low capacity districts. For regressions using the existence variable, I also include bank fixed effects in addition to firm and district-year fixed effects. Regressions using the aggregate variable includes only firm and district-year fixed effects. Col 3-6 present the triple difference event study results by firms classified as high default risk or high MRPK, respectively, and the underlying judicial capacities of the district. Standard errors are clustered by district.

Table A.5: Bankruptcy Reform and District Judicial Capacity: All Firms

	(1) Arc Sine Borrowing	(2) Arc Sine Net Liquidity	(3) Arc Sine Legal Expenditure	(4) Arc Sine Interest Expenditure	(5) Arc Sine Investment	(6) Arc Sine Sales Revenue
High Jdg Pop Ratio x Post	0.0967*** (0.0348)	-0.134 (0.106)	0.0379** (0.0185)	0.0953*** (0.0236)	-0.0231 (0.0227)	-0.0122 (0.0218)
Observations	59025	59070	51405	58395	45825	58560
No. Districts	240	240	228	239	215	239
Firm FE	X	X	X	X	X	X
Industry-Year FE	X	X	X	X	X	X
Mean Dep Var	7.400	2.360	2.870	5.050	5.330	8.650
SD Dep Var	2.200	6.520	1.720	1.830	2.920	1.870
Adj R-Squared	0.803	0.626	0.906	0.803	0.878	0.872

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from DiD specification using time periods before and after the 2016 reform across high and low-judicial-capacity districts on borrowing and production outcomes of all firms in the sample districts. High-judge-to-population ratio districts are those with the ratio above the sample average. Standard errors are clustered by district.

Table A.6: MRPK Estimation Robustness

	(1) Arc Sine Borrowing	(2) Arc Sine Borrowing	(3) Arc Sine Net Liquidity	(4) Arc Sine Net Liquidity
High Jdg Pop Ratio x High MRPK (Within Ind) x Post	0.0201 (0.100)		0.112 (0.170)	
High MRPK (Within Ind) x Post	0.0526 (0.0381)		-0.444** (0.202)	
High Jdg Pop Ratio x Pooled High MRPK (Within Ind Alpha) x Post		-0.0537 (0.0738)		0.511* (0.276)
Pooled High MRPK (Within Ind Alpha)x Post		0.122** (0.0526)		-0.656** (0.259)
Observations	47970	47970	48000	48000
No. Districts	150	150	150	150
Firm FE	X	X	X	X
District FE	X	X	X	X
State-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	7.480	7.480	2.060	2.060
SD Dep Var	2.290	2.290	6.700	6.700
Adj R-Squared	0.797	0.797	0.616	0.616

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above presents robustness using industry-specific estimated capital share $\hat{\alpha}_i$ under Cobb Douglas production technology rather than using a single $\hat{\alpha}$ and classification into high vs. low MRPK within industry or pooled across industries. “Post” is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. High-judge-to-population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above the (industry) median of the firm-level sample. Standard errors are clustered by district.

Table A.7: Using Alternate Cut-off For High Judicial Capacity: Firm Borrowing and Liquidity

	(1) Arc Sine Borrowing	(2) Arc Sine Net Liquidity	(3) Arc Sine Borrowing	(4) Arc Sine Net Liquidity
Above Median Jdg-Pop x High Default Risk X Post	-0.213** (0.0934)	0.471 (0.429)		
High Default Risk X Post	0.135* (0.0710)	-0.567** (0.242)		
Above Median Jdg-Pop x High MRPK X Post			0.0115 (0.0579)	0.485* (0.270)
High MRPK x Post			0.0521 (0.0551)	-0.655** (0.264)
Observations	57840	57885	47970	48000
No. Districts	164	164	150	150
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	7.400	2.360	7.480	2.060
SD Dep Var	2.200	6.520	2.290	6.700
Adj R-Squared	0.801	0.624	0.797	0.616

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use median as the cut-off instead of average. Standard errors are clustered by district.

Table A.8: Using Alternate Cut-off For High Judicial Capacity: Production Outcomes of High Risk Firms

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
Above Median Jdg-Pop x High Default Risk X Post	-0.0460 (0.0555)	-0.147* (0.0785)	-0.254 (0.155)	-0.0290 (0.0916)
High Default Risk X Post	-0.0392 (0.0493)	0.0352 (0.0553)	0.0294 (0.103)	-0.0515 (0.0650)
Observations	50190	57210	44700	57360
No. Districts	152	164	145	163
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.870	5.050	5.330	8.650
SD Dep Var	1.720	1.830	2.920	1.870
Adj R-Squared	0.906	0.805	0.878	0.871

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use median as the cut-off instead of average. Standard errors are clustered by district.

Table A.9: Using Alternate Cut-off For High Judicial Capacity: Production Outcomes of High MRPK Firms

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
Above Median Jdg-Pop x High MRPK X Post	-0.0148 (0.0594)	0.0292 (0.0631)	0.244** (0.119)	0.0585 (0.0506)
High MRPK x Post	-0.0156 (0.0303)	-0.0259 (0.0506)	-0.0588 (0.0972)	-0.159*** (0.0437)
Observations	41070	47400	36795	47520
No. Districts	136	150	130	149
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.910	5.140	5.540	8.690
SD Dep Var	1.760	1.890	2.950	1.940
Adj R-Squared	0.905	0.801	0.877	0.873

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use median as the cut-off instead of average. Standard errors are clustered by district.

Table A.10: Using Pre-Reform Judge Vacancy: Firm Borrowing and Liquidity

	(1) Arc Sine Borrowing	(2) Arc Sine Net Liquidity	(3) Arc Sine Borrowing	(4) Arc Sine Net Liquidity
Below Median Vacancy x High Default Risk X Post	-0.107 (0.117)	0.398 (0.342)		
High Default Risk X Post	0.0938 (0.0759)	-0.566** (0.232)		
Below Median Vacancy x High MRPK X Post			0.0202 (0.0648)	0.255 (0.269)
High MRPK x Post			0.0506 (0.0566)	-0.536*** (0.205)
Observations	57600	57645	47730	47760
No. Districts	162	162	148	148
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	7.400	2.360	7.480	2.060
SD Dep Var	2.200	6.520	2.290	6.700
Adj R-Squared	0.801	0.624	0.796	0.616

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use below median pre-reform judge vacancy rate as the cut-off instead of judge-to-population ratio. Standard errors are clustered by district.

Table A.11: Using Pre-Reform Judge Vacancy: Production Outcomes of High Risk Firms

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
High Default Risk x Post	-0.0525 (0.0480)	-0.0299 (0.0601)	-0.0630 (0.154)	-0.0774 (0.0607)
Below Median Vacancy x High Default Risk x Post	-0.0185 (0.0585)	-0.00775 (0.0968)	-0.0477 (0.148)	0.0242 (0.0891)
Observations	49995	56970	44490	57120
No. Districts	150	162	143	161
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.870	5.050	5.330	8.650
SD Dep Var	1.720	1.830	2.920	1.870
Adj R-Squared	0.906	0.805	0.878	0.871

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use below median pre-reform judge vacancy rate as the cut-off instead of judge-to-population ratio. Standard errors are clustered by district.

Table A.12: Using Pre-Reform Judge Vacancy: Production Outcomes of High MRPK Firms

	(1) Arc Sine Legal Exp	(2) Arc Sine Interest Exp	(3) Arc Sine Investment	(4) Arc Sine Sales Revenue
High MRPK x Post	-0.0104 (0.0429)	-0.0323 (0.0390)	0.0783 (0.0973)	-0.175*** (0.0423)
Below Median Vacancy x High MRPK X Post	-0.0229 (0.0642)	0.0490 (0.0605)	-0.0277 (0.143)	0.0875** (0.0437)
Observations	40875	47160	36585	47280
No. Districts	134	148	128	147
Firm FE	X	X	X	X
District-Year FE	X	X	X	X
Industry-Year FE	X	X	X	X
Mean Dep Var	2.910	5.140	5.540	8.690
SD Dep Var	1.760	1.890	2.950	1.940
Adj R-Squared	0.906	0.801	0.877	0.873

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using a different definition of high judicial capacity. Specifically, I use below median pre-reform judge vacancy rate as the cut-off instead of judge-to-population ratio. Standard errors are clustered by district.

Table A.13: Firm Default Risk and Court Trials

	(1)	(2)	(3)	(4)	(5)	(6)
	No. Filed Defendant	No. Resolved Defendant	No. Pending Defendant	No. Filed Plaintiff	No. Resolved Plaintiff	No. Pending Plaintiff
High Jdg Pop Ratio x High Default Risk x Post	1.008 (0.614)	0.0107 (1.143)	2.855* (1.566)	-0.104 (0.174)	0.219 (0.238)	-0.285 (0.294)
High Jdg Pop Ratio x Post	-0.846 (0.551)	0.456 (0.816)	-2.551* (1.514)	-0.202 (0.350)	0.0467 (0.529)	-0.587 (0.622)
High Default Risk x Post	-0.334 (0.250)	-0.238 (0.351)	-0.356 (0.401)	0.0828 (0.137)	-0.237 (0.226)	0.248 (0.227)
Observations	17037	17037	17037	47340	47340	47340
No. Districts	146	146	146	234	234	234
Firm Fixed Effect	X	X	X	X	X	X
District-Year FE	X	X	X	X	X	X
Industry-Year FE	X	X	X	X	X	X
Mean Dep Var	0.530	0.140	0.390	0.170	0.0200	0.160
SD Dep Var	3.560	1.860	2.540	3.760	0.340	3.540
Adj R-Squared	0.0560	0.0000128	0.120	0.0847	0.0786	0.131

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from Equation 1 using legal case-level outcomes in the corresponding district court as dependent variables for firms classified as high default risk in high-judge-population-ratio districts. “Post” is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes.

High-judge-to-population ratio districts are those with the ratio above the sample average.

High-default-risk firms are those receiving low credit rating (under 4 on a scale 1-8, which map to standard ratings by international rating agencies like S&P or Moody’s) in the period prior to the bankruptcy reform. Col 1-3 present the estimates when such firms appear as defendants in a case, i.e., when someone else files a suit against them. Col 4-6 presents the estimates on case-level outcomes when these firms appear as plaintiff in a case, i.e, when such firms initiate lawsuits in courts. Since the outcome data is at the district court-firm-case-level and that the same firm could have cases in multiple courts, I can identify the coefficient on high judge-population districts interacted with post-reform dummy variable (which would otherwise be collinear with district-year fixed effects). Standard errors are clustered by district. The event study estimates are reported in Figure 6.

Table A.14: Firm MRPK and Court Trials

	(1)	(2)	(3)	(4)	(5)	(6)
	No. Filed Defendant	No. Resolved Defendant	No. Pending Defendant	No. Filed Plaintiff	No. Resolved Plaintiff	No. Pending Plaintiff
High Jdg Pop Ratio x High MRPK x Post	-0.0496 (0.279)	-0.519 (0.545)	0.0687 (0.764)	-0.228 (0.188)	0.389 (0.303)	-0.506 (0.315)
High Jdg Pop Ratio x Post	-1.051** (0.519)	0.666 (0.885)	-2.851** (1.425)	-0.282 (0.367)	-0.322 (0.537)	-0.642 (0.600)
High MRPK x Post	-0.875 (0.575)	0.662 (0.744)	-2.518* (1.280)	0.341 (0.302)	-0.214 (0.273)	0.625 (0.558)
Observations	15291	15291	15291	31761	31761	31761
No. Districts	144	144	144	213	213	213
Firm Fixed Effect	X	X	X	X	X	X
District-Year FE	X	X	X	X	X	X
Industry-Year FE	X	X	X	X	X	X
Mean Dep Var	0.440	0.0900	0.350	0.230	0.0200	0.210
SD Dep Var	3.100	0.890	2.530	4.520	0.390	4.260
Adj R-Squared	0.0692	-0.00140	0.129	0.0688	0.0630	0.110

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates from [Equation 1](#) using legal case-level outcomes in the corresponding district court as dependent variables for firms classified as high-MRPK in high judge-population ratio districts. “Post” is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes. High-judge-to-population ratio districts are those with the ratio above the sample average. High-MRPK firms are those with estimated pre-reform MRPK above median of the firm-level sample. Col 1-3 present the estimates when such firms appear as defendants in a case, i.e., when someone else files a suit against them. Col 4-6 presents the estimates on case-level outcomes when high-MRPK firms appear as plaintiff in a case. That is, when such firms initiate the suit in courts. Since the outcome data is at the district court-firm-case-level and that the same firm could have cases in multiple courts, I can identify the coefficient on high judge-population districts interacted with post-reform dummy variable (which would otherwise be collinear with district-year fixed effects). Standard errors are clustered by district. The event study estimates are reported in [Figure A.11](#).

Table A.15: Banks and Court Trials

	(1) No. Filed Bank Defendant	(2) No. Resolved Bank Defendant	(3) No. Pending Bank Defendant	(4) No. Filed Bank Plaintiff	(5) No. Resolved Bank Plaintiff	(6) No. Pending Bank Plaintiff
High Jdg Pop Ratio x Post	-0.0382 (0.0253)	-0.124*** (0.0369)	0.0281 (0.0447)	0.0907 (0.102)	0.135 (0.222)	-0.125 (0.0821)
Observations	26073	26073	26073	28008	28008	28008
No. Districts	104	104	104	104	104	104
Bank Fixed Effect	X	X	X	X	X	X
District Fixed Effect	X	X	X	X	X	X
Mean Dep Var	1.560	0.250	1.310	1.930	0.290	1.640
SD Dep Var	9.240	2.860	7.380	17.27	5.910	13.52
Adj R-Squared	0.264	0.248	0.256	0.0529	0.0749	0.109

Standard errors in parentheses

* $p < 0.1$, ** $p < .05$, *** $p < 0.01$

Notes: The table above present the estimates using legal case-level outcomes in the corresponding district court as dependent variables for cases involving banks. “Post” is defined as years after 2016, year after the bankruptcy bill becomes an act of the Indian Parliament. Note that the judicial case-level data is only available until 2018, so there are fewer post reform time periods compared to the analysis using firm-level outcomes. High-judge-to-population ratio districts are those with the ratio above the sample average. Col 1-3 present the estimates when banks appear as defendants in a case, i.e., when someone else files a suit against them, such as to prevent liquidation. Col 4-6 presents the estimates on case-level outcomes when banks appear as plaintiff in a case, for example, when they need to enforce a bankruptcy order. Standard errors are clustered by district.